PAPAV: A Capability-Centric Survey of Multimodal Embodied Agents

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Project Page | Awesome Multimodal Embodied Agents

ABSTRACT

Recent progress in multimodal agents and robotics is bringing together two research traditions with complementary strengths. Multimodal agents increasingly sustain long-horizon interaction through reasoning, memory, tool use, and feedback, while robotics has long studied how to close sensing–action loops under physical dynamics. Emerging multimodal embodied agents (MMEAs) must do both, yet existing surveys typically organize these systems through the categories of one tradition and therefore do not directly explain what changes when the same interaction loop is realized through the physical world. This survey addresses this gap by asking what additional responsibilities arise when a multimodal agent must both close and sustain its interaction loop through physical interaction. We introduce PAPAV, which comprises **P**erceive, **A**nticipate, **P**lan, **A**ct, and **V**erify, as a capability-centric coordinate for characterizing the information each function contributes to a recurring task loop, independent of architecture, model family, task, or embodiment. Using PAPAV, we compare multimodal agents, robotic systems, and MMEAs, and analyze how partial observability, limited simulatability, irreversibility, real-time coupling, and unverifiability alter the realization of each capability. Our analysis shows that physical embodiment shifts responsibility toward the running system: agents must maintain persistent physical beliefs, construct action-conditioned futures under uncertainty, formulate and revise task constraints, manage body-specific commitments in physical time, and actively acquire evidence of task outcomes. We further examine representative benchmarks through capability coverage and diagnosticity, revealing where end-to-end success can conceal capability-level failures or leave important parts of the interaction loop under-evaluated. Together, this capability-centric view provides a common basis for understanding, designing, and evaluating agents whose decisions must remain coherent as they pass from multimodal reasoning into consequential physical interaction.

1 Introduction

Recent progress in multimodal foundation models is bringing together two research traditions that have developed with different priorities. Multimodal agents have moved beyond one-shot generation toward systems that combine multimodal perception, reasoning, memory, tool use, and feedback over extended interactions [168, 171, 185, 194, 207, 209, 234]. At the same time, robotic systems are increasingly incorporating foundation models and generalist policies for semantic perception, instruction following, planning, and action generation [3, 11–13, 31, 76, 82, 107, 146, 148].

The two literatures also tend to organize systems from within their own traditions. Surveys centered on multimodal, GUI, and general agent systems commonly organize the literature around agent architectures and components, reasoning and planning, memory and tool use, collaboration, evaluation, and application domains [15, 34, 140, 180, 193, 208]. Surveys centered on embodied AI, VLA models, and robot autonomy more often foreground embodied tasks and platforms, perception, reasoning, low-level control, task planning, execution, integration with robot subsystems, and sim-to-real transfer [118, 132, 163, 184]. Recent surveys have broadened these perspectives, including multimodal-agent frameworks that cover both robotics and GUI/Web applications [140] and embodied-robot frameworks that organize systems around perception, reasoning, execution, and collaboration [184]. However, covering both application areas or proposing a broad functional decomposition is different from placing digital agents and robotic systems side by side and asking how the same function changes between them. Existing categories therefore do not directly explain, across digital and physical settings, how a system builds a view of the current state, anticipates possible outcomes, selects a course, turns that choice into action, and verifies the result, or how responsibility for each function shifts. This limitation suggests a different comparison: rather than matching field labels, we compare how each tradition *realizes and maintains the same interaction loop*.

From this perspective, the two traditions offer complementary strengths. Robotics has long studied how to *close the loop* between sensing and action under physical dynamics [16, 35, 40, 45, 64, 86, 87]. In contrast, recent multimodal-agent research increasingly studies how to *sustain the loop* through context, reasoning, memory, tool use, and feedback over longer interactions [62, 94, 168, 171, 185, 194, 207, 209, 234]. Emerging multimodal embodied agents (MMEAs) must do both. This gap motivates the central question of this survey:

> *What changes when a multimodal agent must both close and sustain its interaction loop through the physical world?*

Motivation for a unified comparison. The above question is timely not because embodied intelligence is new, but because the components needed to close and sustain an interaction loop are becoming more capable, reusable, and tightly coupled. Robotics has long studied state estimation, planning, control, and feedback as core parts of physical interaction [16, 35, 40, 45, 64, 86, 87]. What has changed is the set of tools available to build more general systems [34, 118, 132, 208]. Multimodal and embodied foundation models can connect language, perception, and action across broader task distributions [3, 11–13, 31, 82, 146]; world models and simulation provide increasingly general mechanisms for predicting and rehearsing interaction [14, 57–59, 165, 204]; and agent methods add mechanisms for interleaving reasoning, action, feedback, memory, and self-correction [133, 168, 171, 185, 207, 209]. Large and diverse robot datasets further support the move toward generalist policies [92, 148]. Together, these advances make the opening question a system-level problem. The challenge is no longer only whether a system can perceive, anticipate, plan, act, or verify in isolation, but how these functions remain coordinated and *how responsibility for them shifts* when the loop passes through a changing physical world. A comparison therefore has to track which information is supplied by the environment, which decisions are fixed at design time, and which judgments the running system must make for itself.

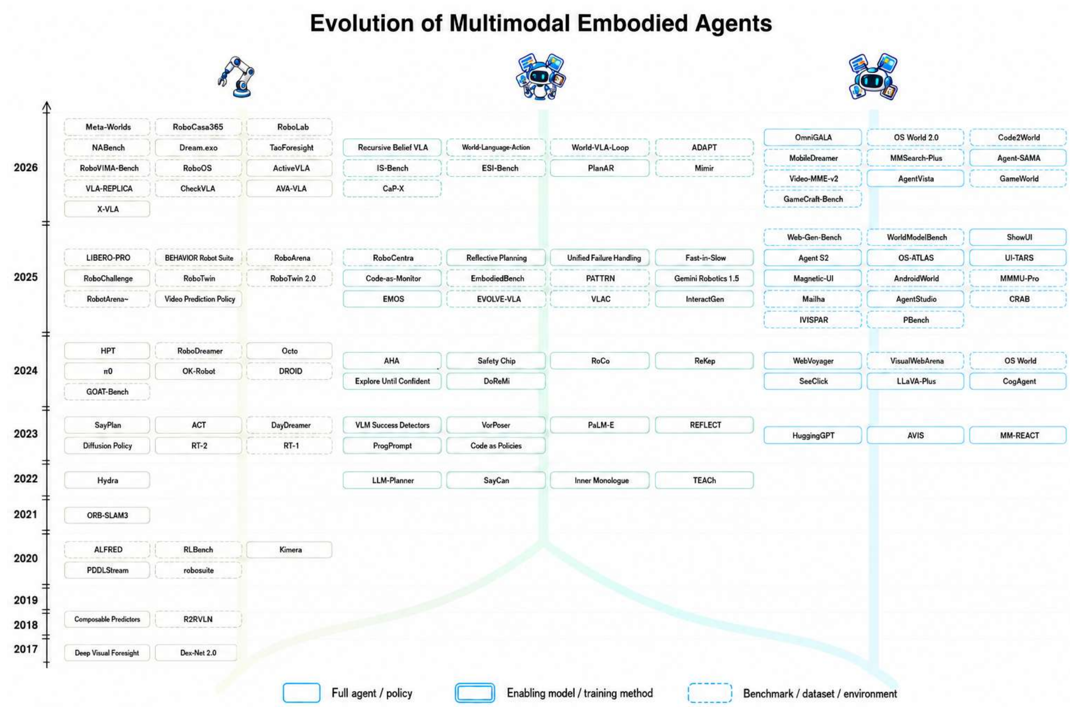

Figure 1. Chronological overview of representative work. From left to right, the three trees summarize robotic systems, multimodal embodied agents, and multimodal agents. Solid, double-bordered, and dashed boxes denote full agents or policies, enabling models or training methods, and benchmarks, datasets, or environments, respectively. The figure includes representative works rather than an exhaustive list. Branches group papers within each area and do not imply direct dependencies between papers.

Figure 1 provides a chronological overview of representative systems, enabling methods, and benchmarks across robotic systems, multimodal embodied agents, and multimodal agents.

A closed-loop capability view. We study this problem through a closed-loop, capability-centric framework. Rather than defining an MMEA by its architecture [34, 193], foundation model [132, 193], task [34, 193], or robot body [118, 132], we organize interaction around five distinct capabilities, each defined by the information it contributes to the loop. **Perceive** produces evidence about the current state. **Anticipate** produces action-conditioned views of possible futures. **Plan** selects a feasible course under the current goal and constraints. **Act** turns that choice into an intervention through a digital or physical interface. **Verify** produces a post-action judgment about what happened and whether more interaction is needed. These capabilities address different questions: what is true now, what may happen, what should be done, how to change the environment, and whether the change succeeded. Each serves a distinct role, and together they cover the main information transformations within a task loop. We call the recurrent loop PAPAV: Perceive, Anticipate, Plan, Act, and Verify. The framework is functional rather than architectural: one model may support several capabilities, while one capability may be distributed across multiple models, memories, planners, tools, and controllers.

The rationale for PAPAV follows from the comparison problem above. Agent taxonomies often organize systems around architectures, reasoning, memory, tools, and digital tasks, whereas robotics taxonomies often organize them around perception, planning, control, tasks, and embodiment [34, 118, 132, 193]. These taxonomies tend to emphasize either computational agent design or embodiment and control. PAPAV instead takes a task-loop perspective, organizing both fields by the functions required to complete an interaction. This view keeps the strengths of both traditions. It captures the agent side by showing how a system maintains a task-relevant state, considers possible consequences, and chooses a course over an extended interaction. It captures the physical side by tracing how that course is grounded in executable action and how the actual outcome is checked before the loop continues. The same five capabilities can therefore describe a digital agent, a conventional robot, or an MMEA, while making visible what changes when an agent's decisions must pass through the physical world.

Its contribution is therefore not a new name for a control loop. Broad closed-loop formulations already capture observation, decision, action, and feedback. PAPAV uses this shared structure as a starting point, but makes two important distinctions explicit: Anticipate separates reasoning about the possible consequences of an action from selecting a course in Plan, while Verify separates observing the resulting state from judging whether the intended outcome was achieved. Using these same distinctions on both sides of the digital–physical boundary helps reveal where familiar assumptions break and what additional responsibilities arise under physical interaction. PAPAV therefore provides a common coordinate for comparing how the loop changes between digital and physical settings, complementing recent agent and embodied-AI taxonomies [118, 140, 163, 184, 208].

This common coordinate also makes the effect of physical embodiment easier to isolate. Digital environments often expose structured interfaces, rereadable state, recoverable transitions, and programmatic success signals [194, 210, 234]. Conventional robotic systems, in contrast, can rely on state representations, planning models, control interfaces, and task conditions that are specified for a known operating regime [16, 45]. An open-ended MMEA can rely fully on neither form of structure. It must operate under **partial observability**, **limited simulatability**, **irreversibility**, **real-time coupling**, and **unverifiability**. These constraints shift more responsibility into the running system. The agent must increasingly maintain a persistent physical belief, construct action-conditioned futures, formulate and revise task constraints, manage body-specific commitments in physical time, and acquire evidence for success or failure. System-level mechanisms for memory, orchestration, timing, uncertainty management, safety, and recovery provide the coordinating substrate, which we refer to as the *agentic harness*. The harness is not an additional capability. Its role is to keep the capabilities coupled as the interaction unfolds.

Position and scope of this survey. Related surveys cover multimodal and GUI agents, embodied agents, robotic foundation models, and VLA systems from complementary perspectives [34, 118, 132, 140, 163, 180, 184, 193, 208]. Some recent surveys span a broad range of multimodal-agent applications, including robotics and GUI/Web settings [140], while others provide functional views of embodied robotic systems [184]. Our focus is different: we place digital agents and robotic systems side by side under the same task-loop coordinates and use that comparison to study what changes when the loop crosses into the physical world.

Table 1 positions related surveys along two scope dimensions, coverage of digital agents and robots, and three analytical dimensions: whether they provide a unified framework across the two, analyze the digital–physical transition, and identify gaps in current benchmarks. Our unit of comparison is therefore the task loop itself rather than a fixed field or system category.

Table 1. Positioning related surveys by scope and analytical focus. For scope, ✓ denotes substantive coverage and × otherwise. For analysis, ✓ denotes an explicit analytical or organizing focus and × otherwise; a cross does not imply that the topic is entirely absent. Full coding evidence is provided in the supplementary claim–evidence matrix.

Survey	Digital agents	Robotic systems	Unified framework	Digital–physical transition	Benchmark gaps
Agent AI [34]	✓	✓	×	✓	×
Large Multimodal Agents [193]	✓	×	×	×	✓
Aligning Cyber Space with Physical World [118]	✓	✓	×	✓	×
(M)LLM-Based GUI Agents [180]	✓	×	×	×	✓
VLA Models for Embodied AI [132]	×	✓	×	×	✓
Towards Embodied Agentic AI [163]	✓	✓	×	×	×
Agentic Multimodal LLMs [208]	✓	✓	×	✓	×
Agentic Artificial Intelligence [15]	✓	✓	✓	×	×
Multimodal Agentic Frameworks [140]	✓	✓	✓	✓	×
LMM-Based Embodied Intelligent Robots [184]	×	✓	×	×	×
PAPAV **(Ours)**	✓	✓	✓	✓	✓

The digital–physical distinction matters because the same task-level functions face different conditions across environments. Digital agents often operate through structured interfaces, recoverable transitions, and programmatic feedback [194, 210, 234], whereas physical systems must contend with sensing uncertainty, dynamics, real-time interaction, irreversible actions, and less direct success signals [13, 31, 45, 118]. We therefore ask not only whether a survey covers both domains, but whether it explains what changes when an interaction loop must be realized through the physical world.

We use binary coding to keep this comparison explicit. For the two scope columns, ✓ denotes substantive coverage and × otherwise. For the three analytical columns, ✓ denotes an explicit analytical or organizing focus and × otherwise; a cross does not imply that the topic is entirely absent. For benchmark gaps, a check requires analysis of limitations or blind spots in current evaluation rather than merely a catalog of benchmarks. The supplementary claim–evidence matrix records the evidence for each coding decision.

Contributions. This survey makes the following contributions:

- **A unified functional framework for digital and physical agents.** We define PAPAV through five capabilities with distinct input–output relations: current-state evidence, action-conditioned futures, course selection, environment intervention, and a post-action verdict. This common coordinate allows digital agents, conventional robotic systems, and multimodal embodied agents to be compared without assuming a particular architecture or treating PAPAV as a replacement for existing control-loop formalisms.
- **An analysis of the digital–physical transition.** We examine how realizing the same interaction loop through the physical world changes the demands placed on its capabilities. Across partial observability, limited simulatability, irreversibility, real-time coupling, and unverifiability, we identify how state maintenance, prediction, problem specification, action commitment, and judgments of success or failure become more demanding and increasingly dependent on decisions made during execution.
- **An analysis of gaps in current benchmarks.** We classify representative benchmarks by whether

each capability is directly diagnosed, only indirectly reflected in an outcome, or not evaluated. This analysis exposes where end-to-end task success can conceal capability-level failures and where current evaluation leaves important parts of the interaction loop underdiagnosed.

Organization. The remainder of the survey first defines the shared task-loop setting and the PAPAV framework. It then compares the five capabilities across digital and physical systems, focusing on what changes as the interaction loop moves into the physical world and on the additional demands introduced by physical constraints. The final parts analyze the coverage and limitations of current benchmarks, identify open evaluation and system-design challenges, and conclude the survey.

2 Background and Problem Formulation

The introduction motivates a comparison between multimodal agents and robotic systems. This section fixes the unit of that comparison. We first summarize the two research traditions, then express both as task-level closed loops, and finally identify the information products that the following *Capabilities: PAPAV section compares*. The formulation is descriptive, not architectural: it does not require a particular model, memory mechanism, planner, or controller.

Throughout this survey, one step means a *task-level intervention*. In a digital environment, an intervention may be a GUI operation, API call, tool invocation, or language act. In a physical environment, it may be a skill call, waypoint, trajectory command, or other semantically meaningful command passed to an execution layer. Individual motor-control updates are inside the execution of that intervention and are not separate PAPAV steps. This choice of level is important: PAPAV compares how systems manage a task, while low-level motion planning and control determine how a physical command is realized.

2.1 Multimodal Agents

A *multimodal agent* is a goal-directed system that repeatedly observes an environment, selects an action, and uses the resulting feedback in later decisions. Its observations may combine text, images, video, audio, screenshots, structured application state, and tool outputs. In the common digital setting, its actions include language generation, clicks, keystrokes, API calls, tool invocations, and code execution. ReAct makes this interaction loop explicit by interleaving reasoning, action, and new observation [209]; MM-REACT and HuggingGPT extend this reasoning–action pattern to multimodal inputs and external tools [168, 207]; and OSWorld evaluates agents through open-ended interaction with real computer applications [194].

The defining property is repeated, state-changing interaction rather than multimodality alone. At task step k, let c_k denote one operation exposed by the interface. It is an environment-facing task intervention, not an internal token or reasoning step. The software environment applies c_k and returns new evidence. An agent may retain that evidence in a full history, compress it into memory, or retrieve it when needed, as in memory-augmented and reflection-based agents [171, 185, 230].

2.2 Robotic Systems

A robotic system closes the same broad interaction loop through a physical body. It senses through hardware and changes the world through actuators. Its observations may include RGB and depth images, LiDAR, proprioception, tactile measurements, force and torque, audio, and inertial signals. Its actions are ultimately realized as joint targets, end-effector motion, locomotion commands, torques, or temporally extended trajectories. Robotics therefore emphasizes state estimation, geometric and dynamic feasibility, and feedback control. ORB-SLAM3 estimates camera motion and map structure from visual–inertial evidence [16]; task and motion planning connects discrete task structure to geometric feasibility [45]; and impedance control regulates physical interaction from feedback [64]. Learned policies such as RT-1 retain this physical loop while replacing parts of the hand-designed stack with learned mappings [12].

The task-level command and the control signal should not be conflated. For a robot with embodiment b, an execution mechanism grounds a task intervention c_k into control inputs over an interval:

$$c_k \xrightarrow{\mathrm{Exec}_b} u(\tau), \qquad \tau \in [\tau_k, \tau_{k+1}). \tag{1}$$

Here c_k may be a selected skill or trajectory request, whereas $u(\tau)$ may be a sequence of joint, torque, or velocity commands. The execution layer must respect embodiment-dependent feasibility and safety constraints. A task decision can therefore be semantically appropriate yet impossible or unsafe for the current body and state. Equation (1) is a separation of analysis levels, not a requirement for two distinct software modules: an end-to-end vision–language–action model may implement the mapping implicitly [11, 13].

2.3 Multimodal Embodied Agents: A Shared Formulation

The two traditions use different interfaces, but both repeatedly estimate what is happening, choose an intervention, observe its consequences, and revise what to do next. We capture only that shared task-level structure.

Let x_k be the task-relevant environment state at intervention boundary k, o_k the evidence available to the agent, g the task goal, and c_k the environment-facing intervention. The interaction history and task belief are

$$h_k = (o_0, c_0, o_1, \ldots, c_{k-1}, o_k), \qquad b_k = B(h_k, g). \tag{2}$$

The symbol b_k means the agent's current task-relevant estimate, including uncertainty where needed. It may be an explicit probability distribution, but need not be. A state estimate, map, scene graph, object memory, or implicit contextual representation can serve the same informational role. Persistent geometric and semantic representations in robotics are concrete examples [16, 50, 77].

The task loop is then

$$c_k \sim \pi(\cdot \mid b_k, g), \qquad c_k \in C_{d,\mathrm{feas}}(b_k), \tag{3}$$

$$x_{k+1} \sim T_d(\cdot \mid x_k, c_k), \qquad o_{k+1} \sim \Omega_d(\cdot \mid x_{k+1}). \tag{4}$$

The domain d determines what these terms mean. In a digital domain, $C_{d,\mathrm{feas}}$ reflects available interface operations, schemas, and permissions, and T_d is largely software-defined. In a physical domain, c_k is grounded through Eq. (1); the resulting transition depends on geometry, dynamics, contact, timing, and disturbances. Thus, T_d summarizes the task-level consequence of executing the intervention rather than replacing the lower-level dynamics.

We use *multimodal embodied agent* (MMEA) for a system in which general multimodal task reasoning is closed around physical execution. The boundary is operationally described by four properties:

1. **Physical state change:** an intervention can change the state of a physical environment, not only a digital representation of it.
2. **Persistent task state:** the system maintains a goal and a task belief across multiple intervention steps.
3. **Semantic capability interface:** task reasoning invokes embodiment-specific execution through semantically meaningful commands or skills.
4. **Closed task feedback:** physical evidence returns to the task loop and can change a later decision, including whether to continue, retry, replan, seek help, or stop.

These properties describe the system boundary used in this survey; they are not an exclusion test for an entire paper. A system may be a modern robotic system and, within a particular task loop, implement MMEA-style task management. Nor do the properties prescribe modularity. PaLM-E integrates continuous embodied observations into a multimodal language model [31]; SayCan combines language-level choices with learned affordances [3]; Inner Monologue returns scene and success feedback to language-model planning [76]; and RT-2 and π_0 map vision and language toward general-purpose robot actions [11, 13].

This formulation now gives a precise bridge from the system-level background to the capability-level comparison: PAPAV names the information transformations needed to keep Eqs. (2)–(4) closed at the task level.

2.4 How PAPAV Appears in the Formulation

PAPAV is not a list of model components. Each capability is defined by the task-level information product it must supply to the loop. A compact typing of the five products is

$$b_k = \text{Perceive}(h_k, g) \in \mathcal{B}_{\text{task}}, \qquad \text{current task belief,} \tag{5}$$

$$\widehat{\Delta}_k(c) = \text{Anticipate}(b_k, c, g) \in \Delta\mathcal{X}_{\text{task}}, \qquad \text{predicted effect of candidate } c, \tag{6}$$

$$p_k = \text{Plan}(b_k, \widehat{\Delta}_k, g) \in \mathcal{P}_{\text{approved}}, \qquad \text{approved course of action,} \tag{7}$$

$$(c_k, \rho_k) = \text{Act}(p_k, b_k) \in \mathcal{C}_{d,\text{exec}} \times \mathcal{R}, \qquad \text{executable command and progress,} \tag{8}$$

$$v_k = \text{Verify}(o_{k+1}, \rho_k, b_k, g) \in \mathcal{V}, \qquad \text{task-consumed verdict.} \tag{9}$$

Here $\widehat{\Delta}_k(c)$ is a task-relevant state change under a candidate intervention, not necessarily a pixel-level rollout or full physical simulation. It may be produced by an explicit world model or by an implicit predictive representation. DreamerV3 and DayDreamer illustrate learned prediction used for decision-making in digital control and physical robot learning, respectively [59, 189].

The set $\mathcal{P}_{\text{approved}}$ distinguishes a proposed option from a course that has passed the system's task, feasibility, or safety checks. Act then exposes executable interaction through c_k and progress through ρ_k; for a robot, the former is grounded into lower-level control by Eq. (1). Finally, $\mathcal{V}$ contains verdicts that alter task-loop behavior, such as *continue*, *success*, *retry*, *replan*, *seek help*, or *fail*. Verification is therefore more than receiving another observation: its output must be consumed when deciding what happens next. Inner Monologue and learned success detectors make this feedback role explicit [33, 76].

Two removal tests clarify why the less obvious roles are separate. Without Anticipate, Plan receives no action-conditioned account of how candidate interventions may change the task state; it can rank actions only by immediate appearance or an equivalent prediction hidden inside planning. Without

Verify, post-action evidence may update the next belief, but the formulation has no explicit judgment that explains why the system stops, retries, replans, or asks for help. An implementation may fuse several PAPAV roles into one model, but it must still provide these information products for the corresponding functions to be present.

2.5 How Physical-World Constraints Change the Loop

Physical embodiment does not create a different task loop; it invalidates defaults that are often convenient in digital interaction. Each constraint below is stated in the same order: the broken default, the PAPAV product that becomes unreliable or unavailable, and the additional runtime responsibility.

Partial observability. *Broken default:* the current interface does not reveal the complete task state. Distinct physical states can produce the same observation, so o_k cannot in general be inverted to recover x_k. *Affected product:* Perceive cannot construct b_k from the latest image alone, and later PAPAV stages inherit this uncertainty. *Runtime responsibility:* the system must fuse evidence across time, maintain persistent state, represent relevant uncertainty, and actively gather observations when the missing information matters. Mapping, scene-graph memory, and active perception are practical responses [16, 50, 77, 122].

Limited simulatability. *Broken default:* a simulator or learned model is not an authoritative transition oracle for open-ended physical interaction. In general,

$$\widehat{T}_b \neq T_b, \tag{10}$$

even when $\widehat{T}_b$ is useful. *Affected product:* Anticipate's $\widehat{\Delta}_k(c)$ is an uncertain prediction rather than transition truth. *Runtime responsibility:* the system must account for model error, monitor execution, preserve contingencies, and revise predictions from real feedback. DayDreamer and UniSim illustrate learned predictive models for physical interaction [189, 204].

Irreversibility. *Broken default:* software-style backtracking cannot undo a physical state change that has already occurred. A feasible intervention need not have a safe, available, or affordable inverse. *Affected product:* Plan's approved course and Act's executable command carry commitment risk; replanning after execution may be too late. *Runtime responsibility:* the system must check preconditions and downstream risk before commitment, authorize conservatively when recovery is limited, and retain abort or recovery strategies during execution [45, 55, 56].

Real-time coupling. *Broken default:* the environment does not pause while the agent reasons, and a physical intervention is not necessarily atomic. *Affected product:* the task belief can become stale during deliberation, while Act's command and progress signal evolve during execution. *Runtime responsibility:* the system must combine slower semantic decisions with faster feedback control, respect deadlines, monitor progress asynchronously, and interrupt or correct execution when necessary. Work on real-time chunking and asynchronous action generation treats this timing mismatch as part of policy design [10, 167].

Unverifiability. *Broken default:* task completion is not a directly readable program state or trusted return value. The system observes evidence of success, not the true task outcome itself. *Affected product:* Verify cannot obtain v_k by simply reading x_{k+1}; its verdict may remain uncertain even after an action finishes. *Runtime responsibility:* the system must acquire task-relevant evidence, use explicit success or failure criteria, express uncertainty, and decide when to inspect again, retry, replan, seek help, or terminate [33, 56, 76].

Together, these constraints explain why the same PAPAV roles demand different information and safeguards in physical settings. They also establish the handoff to the following capability analysis: each comparison can now identify which task-level product changes, why it changes, and what the running system must do in response.

3 Capabilities: PAPAV

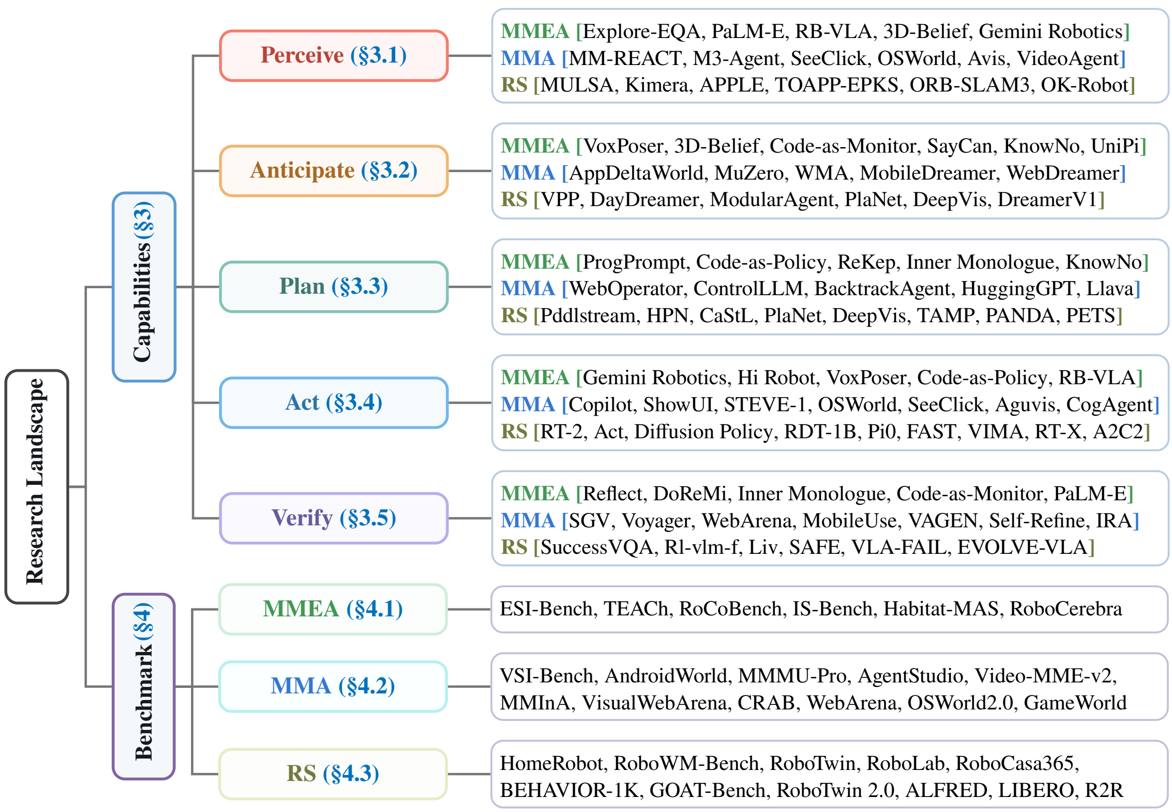

Figure 2. Research landscape and taxonomy of PAPAV. The representative works reviewed in this survey are organized across multimodal embodied agents (MMEAs), multimodal agents (MMAs), and robotic systems (RSs), together with representative benchmarks for each system class.

Multimodal agents, robotic systems, and multimodal embodied agents are complex systems that combine multiple heterogeneous processes, each characterized by different inputs, outputs, representations, and operating conditions. We abstract recurring input–output relations into capabilities and use them as the primary units of analysis. This capability-centred perspective provides a common basis for identifying what these systems share while preserving differences in how each capability is realized and constrained. In particular, multimodal agents and robotic systems may exhibit similar functional relations despite operating in different computational and physical settings, while multimodal embodied agents bring these traditions together: relative to multimodal agents, their capabilities are subject to embodiment and physical-world constraints; relative to conventional robotic systems, they incorporate stronger agentic properties such as multimodal reasoning, goal-directed autonomy, and adaptive interaction.

We select **Perceive, Anticipate, Plan, Act, and Verify (PAPAV)** as five analytical coordinates because they provide a minimal yet sufficiently expressive decomposition for the scope of this survey. The five capabilities are distinguished by qualitatively different input–output relations and therefore separate analytical questions that would be obscured if combined. At the same time, they collectively span the principal forms of transformation through which goal-directed systems derive, organize, enact, and assess interaction-relevant information. PAPAV is therefore not intended as an exhaustive inventory of everything an intelligent system may possess, nor does it prescribe five architectural modules, a fixed execution order, or a universal processing pipeline. Instead, it provides an architecture-independent and cross-domain coordinate system at a granularity that preserves meaningful distinctions while supporting consistent comparison.

Each following section accordingly takes one PAPAV capability as its analytical coordinate, first identifying similarity between multimodal agents and robotic systems and then examining how the corresponding capability is inherited, coupled, or reconfigured in multimodal embodied agents.

3.1 Perceive

Purpose and mechanism. Perception constructs an action-relevant representation of the current environment, task, and agent state from heterogeneous, partial observations. Across multimodal agents and robotic systems, Its central operation is *state grounding and estimation*: mapping raw observations and interaction history to a structured or latent belief over task-relevant entities, relations, affordances, and states. This step is the bridge from sensing to agency. Recognizing all visible content is neither necessary nor sufficient; the representation must expose what can be acted upon and what matters for the current objective.

Physical embodiment preserves this functional role but changes who must resolve it, when, and from what evidence. Under partial observability the action-relevant belief is no longer a transient mapping from one observation to one representation; it becomes an artifact the agent maintains. Object-centric memories and agent-editable scene graphs retain state beyond the current view, and active-perception policies fold observation selection into the policy itself [4, 50, 122]. A multimodal embodied agent (MMEA) must then preserve identity across viewpoints, answer open-vocabulary queries, record uncertainty and provenance, and revise or invalidate earlier entries as it acts. Verify asks something narrower: whether goal-specific evidence supports one particular outcome judgment.

3.1.1 Perceive: Similarity

From Incomplete Observations across Individual Modalities to Integrated Multimodal Evidence. Individual modalities capture different aspects of a situation and have different limitations: images provide appearance and spatial cues, audio and video reveal events and temporal changes, while depth, proprioception, force, and touch provide information about geometry, embodiment, and physical contact. Therefore, Both multimodal agents and robotic systems combine complementary information from multiple modalities to construct a more reliable representation of their environment. Multimodal agents may combine image regions, text, audio, video segments, interface states, and outputs returned by specialized tools, whereas robotic systems may integrate RGB images, depth, proprioception, force, touch, and geometric cues. Depending on the task, this process can also involve spatial or temporal alignment, the reconciliation of inconsistent cues, and the use of contextual or historical information [97, 102, 123, 168, 207]. MM-REACT, for example, combines image and video evidence obtained from specialized visual tools, while M3-Agent organizes visual and auditory observations in a shared entity-centered memory [123, 207]. Similarly, See, Hear, and Feel integrates global visual context, acoustically detected events, and local tactile geometry for robotic manipulation [102]. Although these systems operate with different inputs and in different environments, they share the same perceptual strategy: combining complementary modalities to recover task-relevant information that is incomplete, ambiguous, or unreliable in any individual source.

From observations to Grounded Environment State Representations. Integrated evidence remains insufficient for downstream reasoning unless it is grounded into an environment state representation that specifies what currently exists and how it is changing. Perception must therefore associate observations with identifiable entities and events, estimate their current attributes, establish relations among them, and preserve their identities across time[50, 123, 151, 160, 187]. The resulting representation may describe objects, speakers, video events, temporal intervals, digital entities, object poses, spatial configurations, motion, contact, or robot state. These variables differ across environments, but they share identity, current state, relational structure, and temporal persistence to predict [50, 77, 152, 162, 179, 181]. SeeClick grounds language in screen coordinates, while OSWorld requires agents to recover application and interaction state from visual observations [25, 194], while Kimera converts continuous visual–inertial measurements into a dynamic scene graph containing persistent objects, agents, places, and spatiotemporal relations [162].

Together, these shared mechanisms show that Perceive is not tied to a particular sensor suite: it integrates available evidence and grounds it into a task-relevant state representation [50, 102, 207]. They do not, however, determine how missing evidence should be sought when the needed state cannot be recovered from current observations. Under embodiment, observation becomes a situated and constrained interaction [164], so the critical question shifts from multimodal fusion alone to how task-level uncertainty is connected to sensorimotor acquisition and subsequent decisions. The following contrasts therefore examine, first, how digital evidence requests become embodied inquiry [70] and, second, how an MMEA differs from task-conditioned robotic active perception [47, 159] in the division of responsibility for forming questions, controlling acquisition, and using the returned evidence.

3.1.2 Perceive: Difference

MMEA versus MMA: From Digital Evidence Requests to Embodied Inquiry. Information seeking already forms part of the MMA decision loop. In AVIS, tool selection follows from earlier results, returned evidence is interpreted by a reasoner, and working memory carries that evidence into subsequent search [70]. Embodiment relocates the inquiry to the agent's own sensorimotor relation with the environment. A pending physical commitment may expose either an explicit question or an unresolved state variable, while viewpoint, contact, timing, energy, and disturbance constrain how the uncertainty can be resolved. The resulting *qualified belief* records the relevant proposition or state estimate together with confidence, provenance, acquisition conditions, and freshness. Its task significance is established when the update changes the plan, the next capability invocation, or the parameters under which physical execution may proceed.

MMEA versus Task-Conditioned Robotic Active Perception: From Supplied Objectives to Task-Formed Questions. Robotic active perception covers both supplied and task-conditioned information needs. APPLE learns acquisition policies across several active-perception problems, Explore until Confident chooses movement and stopping for a supplied embodied question, and task-oriented formulations revise plans as observations change beliefs over task propositions [47, 159, 164]. PAPAV places the Perceive boundary at the ensuing division of responsibility. The evolving task state formulates or revises the semantic evidence request and later uses the qualified return to choose among capabilities and physical commitments; the sensing specialist retains control of camera poses, tactile trajectories, stopping, and acquisition cost. Any robotic implementation that exposes this question–acquisition–decision trace satisfies the capability relation, while whole-system classification follows the joint rule developed in Section 3.6.

3.1.3 Perceive: Summary

Perceive turns a task-level information gap into a decision-bearing belief revision before physical commitment. Its minimum observable record comprises the information need, the embodied acquisition, the qualified return, and the decision to which that return is applied. Belief calibration, evidence freshness, acquisition cost, and the frequency of evidence-induced revisions can be measured from execution traces. Value of information requires paired trials, branching replay, or a validated simulator because it depends on the unobserved course of action without the acquired evidence. The capability thus binds physical sensing to the particular commitment that made the evidence necessary.

3.2 Anticipate

Purpose and mechanism. Anticipation reduces uncertainty about how the environment may evolve and what consequences candidate actions may produce before commitment. We define it as *action-conditioned prospective modeling*: given a current belief b_t and a candidate action a_t, the system estimates a distribution over future states, observations, rewards, risks, or task progress. The defining question is therefore counterfactual—“what would follow if this action were taken?”—rather than merely predictive—“what is likely to happen next?”

Anticipation keeps this prospective, action-conditioned character under embodiment, but limited simulatability denies a multimodal embodied agent (MMEA) a complete and authoritative simulator for open-ended physical interaction. Existing systems predict pixels or multi-view video, latent tokens, object state, contact, force, and task progress [88, 110, 177, 217, 226, 237]. Which representation is chosen matters less than the criterion it must meet: a forecast is useful only if it discriminates among candidate actions and thereby alters planning or execution. Visual plausibility alone does not establish embodied realizability for the commanded action [205].

3.2.1 Anticipate: Similarity

Digital and physical environments instantiate this function at different levels of abstraction. For multimodal agents, a candidate action may invoke a text-to-image model, describe post-action webpage, synthesize speech or music, generate a video, call a software tool [18, 37, 52]; for robotic systems, it may change future poses, velocities, contacts, collisions, object motions, rewards, and termination [17, 35, 40, 87, 189]. However, Both systems nevertheless require a prospective account of how their available actions may change a heterogeneous multimodal environment.

From Current State Representations to Predicted Environmental Change through Predictive Models. Anticipate provides Plan with information that cannot be obtained from the current environment state alone by modeling how the environment may change over time, under different contexts, or in response to possible external interventions. Starting from the environment state representation constructed by Perceive, both multimodal agents and robotic systems use predictive models to form probabilistic or structured expectations over possible future states. These models may take the form of learned transition models, generative observation models, latent dynamics, structured predictors, analytic or hybrid simulators, and tool or environment emulators [14, 52, 59, 112, 189, 192, 204]. In multimodal agents, predictive models can capture possible changes in language-described environments, video states, webpages, GUI applications, and other interactive digital environments [17, 37, 52]. In robotic systems, they can predict how control inputs may change future visual observations, object configurations, robot states, and task outcomes [35, 69, 87]. A world model is a general form of predictive model that represents reusable environment dynamics. Compared with a conventional predictor designed to estimate a fixed target for a specific task, a world model can recursively project multiple future steps and condition those projections on different contexts or hypothetical interventions, thereby supporting a wider range of prospective queries [57, 112, 165, 192]. Its advantage therefore lies in the generality and reusability of its dynamics representation, rather than in guaranteed predictive accuracy. Similarly, a generative model functions as a world model only when its output is treated as a possible successor state for reasoning, rather than being directly committed to the environment. The output of Anticipate is thus a predictive prior or a set of plausible future states; ranking candidate actions on the basis of those futures belongs to Plan.

From Predicted Change to Semantic or Latent States. A useful predictive model need not expose every pixel, waveform, or sensor reading generated during a rollout. Images, videos, and multisensory trajectories may serve as internal prediction mechanisms, but passing all of their raw details to downstream capabilities is computationally costly and may obscure the changes that matter for the current task [18, 58, 61, 165, 200, 221]. Both multimodal agents and robotic systems therefore abstract complex predicted evolution into semantic or latent future-state representations. A semantic future state explicitly describes which entities, events, relations, object states, affordances, constraints, or indicators of task progress are expected to appear or change. It may also encode these predictions as a state delta relative to the current environment state [3, 52, 200, 232]. A latent future state instead uses compact learned variables to represent environment dynamics and possible outcomes without reconstructing every observable detail [57, 69, 165]. These representations may encode a single future, multiple plausible futures, or uncertainty over future outcomes, and semantic and latent variables can also be combined. AppDeltaWorld represents changes in mobile GUI environments as transition-grounded delta code, providing an explicit description of predicted state changes rather than reproducing complete future interfaces [200]. MuZero learns latent dynamics that preserve information relevant to rewards, policies, and values without reconstructing observations, while Video Prediction Policy extracts predictive visual representations for robotic control [69, 165]. Across both digital and physical environments, Anticipate preserves the information needed to characterize possible environmental change.

The shared mechanisms convert a grounded current state into task-relevant representations of how candidate interventions could change the environment, using predictive models and semantic or latent abstraction [58, 200]. In open-ended physical interaction, however, a plausible future is not yet a decision-worthy forecast: simulator inaccuracies, contact uncertainty, and environmental change can make it informative in one setting and misleading in another [189, 205, 226]. Existing robotic prediction ranges from control-internal imagination to task-aware semantic–dynamic coupling [189, 219]; embodiment therefore requires us to ask whether the counterfactual is selected and consumed at the task level. The following contrasts examine how MMEAs qualify physical forecasts relative to digital futures and how they connect task-selected counterfactuals to predictive mechanisms relative to control-centric robotic prediction.

3.2.2 Anticipate: Difference

MMEA versus MMA: From Inspectable Digital Futures to Qualified Physical Forecasts. Action-conditioned prediction supports candidate comparison in digital as well as embodied settings. Web Agents with World Models estimates the next observation and reward for web actions and uses those estimates in policy selection [18]; the software platform and its external services delimit the transitions for which the model is informative. Physical forecasts instead depend on estimated dynamics, contact, embodiment, and disturbances that remain only partly controlled at deployment. Their decision value is delimited by a *validated applicability envelope*: the predicted variables and horizon, the operating conditions represented during evaluation, and the signals used to recognize departures from those conditions. A qualified consequence couples the predicted state change or outcome distribution with its uncertainty, applicability envelope, and unsupported-forecast indicator, so that Plan receives the limits of the forecast together with the forecast itself.

MMEA versus Control-Centric Robotic Prediction: Task-Level Selection of Physical Counterfactuals. Robotic world models range from control-oriented prediction to task-aware semantic–dynamic coupling. DayDreamer uses imagined action outcomes for policy improvement while relying on real interaction to accommodate simulation error and unmodeled change; ModularAgent exchanges information bidirectionally between semantic reasoning and latent dynamics [189, 219]. In the PAPAV account, Anticipate is identified by a functional interface rather than a prescribed architecture. The current task decision specifies the physical counterfactual of interest, the predictive mechanism allocates an appropriate model, horizon, precision, and computational budget, and a qualified consequence returns to the process that retains commitment authority. Modular implementations may expose this exchange directly. For end-to-end policies, prediction-branch ablations, latent interventions, or induced changes in candidate ranking can supply the observable evidence needed for capability-level attribution.

3.2.3 Anticipate: Summary

Anticipate supplies Plan with qualified consequences for task-selected physical counterfactuals. Calibration on executed transitions and detection of conditions outside the validated applicability envelope are directly observable; ranking agreement and decision regret additionally require outcomes for actions that were not executed and thus depend on branching simulation, replay, paired trials, or repeated physical evaluation. Representational adequacy is task-relative: a compact state delta may outperform a high-fidelity rollout when it preserves the consequence differences that govern commitment and marks the limits of its own validity.

3.3 Plan

Purpose and mechanism. Planning selects a course of action expected to advance the goal under the current belief, anticipated outcomes, and applicable constraints. Its core pattern is *constrained proposal and selection*: semantic models generate plausible subgoals or action sequences, while tool schemas, affordance scores, search procedures, or external verifiers determine which proposals are executable. Language Models as Zero-Shot Planners grounds free-form steps in an admissible action set, SayCan combines language-model scores with learned affordances, and LLM-Modulo places model-generated proposals inside a verifier-guided refinement loop [3, 73, 89].

Irreversibility changes the planning objective without changing this function. An MMEA must convert an open-ended multimodal goal into goal conditions, objects, constraints, costs, subgoals, and skill choices, and then decide whether the resulting plan warrants execution [55, 75, 158]. Planning under physical commitment thus comprises problem construction followed by commitment authorization, where clarification, abstention, assistance, and escalation count as legitimate outcomes and not as failures to select an action. Act, in turn, governs a question that arises only afterwards: how an authorized commitment is realized and revised in physical time.

3.3.1 Plan: Similarity

From Complex Task Goals to Hierarchically Structured Subgoals. Planning commonly begins from a task goal, a representation of the current state, predicted consequences, and the capabilities available in the environment [3, 44, 52]. Because long-horizon tasks contain complex dependencies and combinatorial choices, both multimodal agents and robotic systems decompose goals into intermediate subgoals, order their dependencies, and progressively refine them into reusable components at finer decision scales [1, 86, 103, 154, 156]. Multimodal agents can construct workflows from GUI operations, tools, APIs, or programs, whereas robotic systems can compose navigation and manipulation stages, motion primitives, and learned skills [107, 116, 154, 170]. ProgPrompt expresses situated task plans as programs, while Code as Policies composes perception and control APIs into executable procedures [107, 173]. Across digital and physical environments, the shared role of abstraction is to express the task goal and environment state against a reusable vocabulary of tools, skills, or programs, while preserving the environment-specific conditions under which each capability is available.

From Candidate Plans to a Selected Course under Constraints. Given this structured problem representation, reliable planning requires deliberation over alternative courses rather than a one-shot continuation. Candidate plans can be searched, scored under predicted outcomes, rejected when preconditions fail, and revised after new evidence. Both system classes evaluate alternative courses according to a combination of goal relevance and current-state feasibility. Multimodal agents may compare predicted webpage outcomes, select specialized multimodal tools, or construct dependency-aware multi-tool workflows [52, 116, 168, 191]. Robotic systems may compare skills, symbolic task sequences, grasps, and trajectories while enforcing affordance, kinematic, collision, and motion constraints [3, 44]. WebOperator uses action-aware tree search to generate and rank web actions, filter invalid candidates, and backtrack from unsafe paths, while ControlLLM searches a tool graph that encodes parameter and dependency relations [30, 121]. ReKep represents a robotic manipulation course as a sequence of relational keypoint constraints and optimizes end-effector poses subject to those constraints [75]. Consequently, a missing API operation and an unreachable grasp play the same functional role: each invalidates a candidate that might otherwise appear relevant to the goal. The resulting output is a coherent and feasible course from the current state toward the goal: it specifies the selected intermediate objectives or interventions, resolves their dependencies and ordering, and satisfies the constraints governing their applicability [65, 121, 168].

These similarities show that both systems can structure long-horizon goals into subgoals and reusable capabilities [86, 168], then select among candidate courses by testing dependency constraints and environment-specific feasibility [3, 121]. This shared account explains how a course becomes coherent and feasible, but not how planning should proceed after execution produces a discrepant state that must be diagnosed and recovered from [56, 120, 190]. Physical commitment therefore makes the consequences of selection—and the recoverability of the resulting state—part of planning itself rather than merely feedback for a later retry. The following contrasts examine recovery from a physically altered state relative to digital plan revision and MMEA task-level governance of commitment and recovery relative to local robotic replanning [76, 158].

3.3.2 Plan: Difference

MMEA versus MMA: Recovery from Physically Altered State. Digital and embodied agents both revise plans after execution feedback, but the state from which recovery begins differs materially. BacktrackAgent detects GUI errors with verifier and judger modules and uses reflection to recover a productive interaction state; the relevant pages and software transitions remain available in the interaction history [190]. A failed physical commitment may instead distribute change across geometry, contact, material condition, tool availability, and consumed resources without leaving a complete transaction record. Subsequent planning must reconstruct the state that is actually recoverable, identify effects that require compensation, and account for options removed by the preceding action. This makes reversibility a planning variable before commitment and throughout recovery from the altered physical state, and it determines which remedies remain admissible under the continuing task objective.

MMEA versus Local Robotic Replanning: Governance of Commitment and Recovery. Robotic replanning already makes productive use of execution feedback. Inner Monologue returns scene and success observations to a planner so that another skill can be selected, whereas KnowNo treats calibrated help-seeking as a valid response to ambiguity [76, 158]. The MMEA planning relation is distinguished by its temporal scope: authorization of the original commitment and selection of the recovery course remain accountable to a persistent task state containing the goal, constraints, decision history, and consequences already incurred. Before execution, that state may warrant further evidence, assistance, or an informative action that preserves alternatives; after failure, it supports a choice among continuation, compensation, renewed perception, escalation, and stopping. A safety controller or human operator may retain veto authority without taking over task planning. Once the human selects the next task action after each outcome, however, the system has a human-in-the-loop planning profile rather than a fully autonomous Plan relation.

3.3.3 Plan: Summary

Plan spans pre-commitment authorization and post-outcome stewardship of the same task. Local skill or motion replanning is one recovery instrument alongside renewed perception, compensation, assistance, and stopping. A physical action reshapes the future decision space: its retained option set contains reachable safe continuations, recovery branches, and compensating actions, while retained option value applies an explicit task-utility and risk model to that set. Risk-weighted precondition coverage and the frequency of uncompensated persistent errors are available from ordinary execution records. Estimating option value or expected recovery cost additionally requires branching rollouts, replay, or a documented reachability approximation. Planning quality depends on the courses preserved after action as well as on the action initially selected.

3.4 Act

Purpose and mechanism. Acting realizes a selected decision as an environment-valid intervention. Action grounding maps an abstract intention to the arguments, coordinates, tokens, trajectories, or control signals accepted by the environment, after which closed-loop execution issues the command and observes its immediate effect. This boundary matters because a semantically correct plan can still fail when the target, command schema, or action parameterization is grounded incorrectly.

Real-time coupling prevents the physical environment from pausing while a semantic model deliberates, so an authorized plan must be realized as a body-specific commitment in physical time. In asynchronous chunked policies, one forward pass emits a trajectory segment while the next inference computes its successor, and the new segment can blend with or correct the unexecuted remainder [10, 167]. The commitment is revisable, not atomic. Its interface must also identify the controlled body, because the same semantic instruction induces different trajectories across morphologies. Recent studies accordingly treat action-space design and action tokenization not as incidental implementation details but as architectural variables in their own right [38, 90].

3.4.1 Act: Similarity

From Semantic Interventions to Grounded Action Specifications. Act receives a semantic intervention from Plan, but the intervention is not executable until its target, primitive, arguments, and reference frame are grounded in the target environment. Both multimodal agents and robotic systems resolve this gap by interpreting the intervention together with the current state representation and the available action space. Multimodal agents may map an instruction such as "click Submit" to a GUI element and screen coordinate, an API call, a typed command, or a code invocation [62, 201, 229]. Robotic systems may map "pick up the cup" to a perceived object, robot arm, coordinate frame, grasp pose, learned skill, or low-level control target [24, 82]. Although these targets and action primitives differ, they serve the same function: determining where and how a intervention can be realized in a particular environment. OS-Copilot grounds configured subtasks into executable operations across web, terminal, file, multimedia, and application interfaces, whereas RT-2 maps visual observations and language instructions directly to robot actions [13, 191]. The resulting output is an environment-specific action specification that can be executed by the relevant interface or controller.

From Grounded Action Specifications to Action Representations Compatible with the Environment. A grounded intervention remains an internal specification until it is decoded into the action representation and temporal granularity required by the target interface. Both multimodal agents and robotic systems therefore transform grounded interventions into environment-compatible actions. Multimodal agents may output screen coordinates, structured GUI operations, typed commands, tool or API calls, executable code, or streams of discrete digital actions [25, 48, 66, 111, 113, 229]. Robotic systems may produce discretized action tokens, action chunks, waypoints, or continuous trajectories [12, 13, 82, 228]. Although these representations differ, they serve the same function: expressing an intended intervention in a form that the corresponding environment can execute. ShowUI generates structured GUI actions containing operations and interface locations, while RT-2 represents robot actions as tokens that can be generated autoregressively [13, 113]. STEVE-1 generates sequences of low-level keyboard and mouse actions over time, whereas ACT predicts short chunks of robot actions rather than isolated commands [111, 228]. For high-dimensional continuous robot control, Diffusion Policy and RDT generate action trajectories or chunks through diffusion, while π_0 uses flow matching [11, 27, 117]. These are specific implementations for robotic control rather than architectures that both system classes must share. As execution unfolds, subsequent actions can be conditioned on updated observations, allowing both multimodal agents and robotic systems to respond to environmental changes.

Taken together, these similarities establish a common execution relation: a selected intervention is grounded to an environment-specific target and action space, encoded at an executable temporal granularity, and can be adjusted as new observations arrive [13, 25, 167]. This account explains how intentions become commands, but leaves open how the task-level meaning of a commitment remains connected to body-specific control as execution unfolds in physical time [10]. The comparison with MMAs therefore asks how a compact agent-facing operation [194] must expand into an interface that carries information both to and from temporally extended physical execution [10]. The comparison with goal-conditioned robotic execution [13, 27, 228] then asks when body-specific control becomes part of an MMEA capability rather than a self-contained motor process—namely, when its execution state remains visible and consequential to the ongoing task process.

3.4.2 Act: Difference

MMEA versus MMA: Extending the Agent-Facing Interface to Physical Execution. MMAs commonly issue compact operations and leave lower-level semantics to a software runtime. OSWorld, for example, presents clicks, text entry, code, and other platform-defined interfaces to the agent [194]. Physical execution extends this interface across embodiment and time: a semantic commitment must be realized by a particular body while remaining available to the task process as execution unfolds. The downward contract identifies the capability, target, parameters, preconditions, and admissible interruption points. In return, the controller reports its state, estimated state change, progress, termination status, low-level faults, and observation time. Together, the two directions preserve the task meaning of the commitment while leaving implementation and timing to embodiment-specific control.

MMEA versus Goal-Conditioned Robot Execution: A Task-Visible Contract above Body-Specific Control. Robotic policies already realize compact objectives on particular bodies. RT-2 tokenizes robot actions, whereas ACT and Diffusion Policy produce temporally extended motor sequences [13, 27, 228]. Within PAPAV, capability attribution turns on the visibility of the execution contract to the current task decision, rather than on the motor policy's internal representation. At minimum, the downward message specifies the target, intended state change, governing constraints, and interruption conditions; the upward message reports controller-estimated change, progress, termination status, faults, and observation time. This telemetry neither substitutes for Perceive's estimate of the physical state nor determines Verify's postcondition verdict. An explicit API, a modular control stack, or an end-to-end policy can instantiate Act, provided that time-sensitive execution feedback can still modify the course while physical options remain available for task-level intervention.

3.4.3 Act: Summary

Act is the continuity relation between an authorized intervention and its body-specific realization. Grounding correctness and interruption latency measure its two principal risks; progress observability, termination-state accuracy, and missed deadlines further characterize the execution contract. Semantic divergence should compare the authorized target, intended state change, and protected constraints with the controller's execution record, rather than compare two verbal descriptions. Specialized control remains compatible with task-level agency when the authorized commitment stays traceable and interruptible throughout physical execution and subsequent recovery.

3.5 Verify

Purpose and mechanism. Verification determines whether an executed intervention produced the intended change, advanced the task, and remained consistent with relevant constraints. Its evidence must come from the post-action environment rather than from the model's confidence in its own proposal. Across multimodal agents and robotic systems, this capability compares task-dependent completion conditions with external evidence, including GUI states, application data, camera observations, proprioception, contact signals, and execution traces. Accordingly, verification is stronger than free-form self-reflection: it grounds judgments of success, failure, and constraint satisfaction in observable consequences of action [29, 95, 169]. Digital benchmarks such as WebArena and OSWorld use observable page, application, or execution state to determine completion; for Robotic Systems, Inner Monologue feeds success signals and scene feedback back into language-model planning, while vision-language success detectors infer outcomes from observations [33, 76, 194, 234].

Embodied verification is still retrospective adjudication after action, but unverifiability removes the independent adjudicator that digital environments frequently supply. The constraint also sharpens the boundary with Perceive: Perceive maintains a task-general belief, whereas Verify tests goal-specific criteria against selected post-action evidence. Learned verifiers increasingly act as coaches rather than referees, selecting among candidate actions, supplying dense progress rewards, supporting test-time adaptation, or converting verified trajectories into training data [9, 96, 218, 222, 223]. That role makes verification operationally useful, and it also places the verifier's own errors inside the very control and learning loops that it is meant to safeguard.

3.5.1 Verify: Similarity

From Observed and Expected Outcomes to Verification Judgments. Verification requires two grounded inputs: evidence of what occurred after an action and an independently specified account of what should have occurred [127, 155]. In digital environments, such evidence may include tool returns, outcome pages, persistent interface states, and interaction histories; in physical environments, it may additionally include visual, proprioceptive, contact, force, and other temporally accumulated observations. These signals allow GUI agents to inspect state transitions and robots to reconstruct manipulation outcomes [43, 53, 120, 190]. The corresponding reference may be a task goal, predicted next state, subgoal condition, or explicit success criterion. Self-Grounded Verification addresses this problem by first eliciting task-completion criteria independently of the candidate trajectory and only then evaluating the trajectory against those criteria; importantly, the same mechanism is demonstrated across Web agents, GUI agents, and robotic manipulation policies [6]. Physical-domain critics additionally require causal, spatial, and affordance awareness, while runtime failure detectors must account for uncertainty and out-of-distribution execution states [195, 196]. Earlier reflection-based methods establish useful feedback–revision loops, but self-generated critique alone does not guarantee correspondence with the actual environment [133, 171]. Verification should therefore be grounded in independently specified criteria, multimodal observations, tool-accessible states, or calibrated anomaly evidence rather than solely in the agent's own explanation.

From Verification Judgments to Corrective Feedback. Given grounded outcome evidence and expected outcomes, verification must convert their comparison into feedback that can regulate execution rather than merely produce a detached binary label. Both multimodal agents and robotic systems need to distinguish terminal success from intermediate progress, recoverable mismatch, insufficient evidence, invalid expected transitions, and persistent risk. A verifier may therefore determine subgoal completion, estimate continuous progress, compare alternative outcomes, or rate an entire trajectory together with its uncertainty. Language-conditioned success classifiers and multimodal reward models instantiate this mechanism by converting observed state changes into signals that evaluate execution and can also support policy learning [130, 131, 143, 186, 188]. In digital environments, a changed application state or failed tool return can reveal whether an action advanced the task. Voyager, for example, combines environment feedback, execution errors, and self-verification to revise subsequent behavior [185]. In physical environments, verification may instead rely on post-manipulation object relations, poses, or contact evidence. REFLECT summarizes robot experience to explain and correct failures, while DoReMi detects and recovers from misalignment between planned and executed behavior[56, 120]. Across both system classes, an actionable judgment should identify not only whether the observed transition matches the expected transition, but also where the mismatch occurred and what type of corrective response it requires.

Verification establishes a closed loop: post-action evidence is compared with independently specified expectations [6, 127], and the resulting judgment is converted into feedback that can regulate subsequent execution [120]. Yet this shared mechanism leaves unresolved how a verdict can remain timely when physical evidence is partial or delayed [2, 119], and reliable when observations are uncertain [196] or the verifier shares evidential dependencies with the acting stack. The comparison with MMAs therefore shifts from software-mediated functional checks [104, 234] to physically grounded task verdicts [195] whose timing and evidential dependencies affect their reliability. The comparison with local robotic outcome monitoring [33, 56] then asks whether a judgment remains a local monitoring signal or is returned for task-level use during the ongoing episode.

3.5.2 Verify: Difference

MMEA versus MMA: From Functional Checks to Physically Grounded Task Verdicts. Digital and embodied verification both compare observed outcomes with task-specific completion conditions. WebArena illustrates external programmatic judgment, whereas MobileUse places hierarchical reflection inside the agent's ongoing task loop [104, 234]. In an MMEA, the judgment rests on post-action physical evidence affected by partial visibility, sensor uncertainty, and delayed consequences. Timeliness becomes part of correctness: a verdict must reach the task process while continuation, renewed perception, replanning, recovery, assistance, or stopping can still change the outcome. Reliability also depends on the verifier's evidential-dependence profile. Shared sensors, representations, model backbones, or training data may produce correlated errors despite architectural separation, so verification records should disclose provenance and dependencies and quantify correlated failures wherever the evaluation design permits comparison with the acting policy.

MMEA versus Local Robotic Outcome Monitoring: Task-Level Consumption of Physical Verdicts. Robotic outcome monitors serve several destinations. Vision-Language Models as Success Detectors turns observed behavior into a reward or evaluation signal, while DoReMi identifies plan–execution misalignment for corrective action [33, 56]. The resulting judgments may govern local termination, revise the current episode, adapt a policy across episodes, or label data offline. Verify acquires its MMEA role when the current task process consumes an actionable verdict before selecting the next course. Such a return states the judged outcome, confidence, supporting evidence, unresolved conditions, and estimated intervention window. A controller stop, a task-level verdict, and an offline label may share a detector while retaining different responsibilities; capability-level attribution follows the route and use of the judgment during the episode.

3.5.3 Verify: Summary

Verify adjudicates whether post-action evidence satisfies the task postcondition while another decision remains possible. Act contributes execution telemetry and low-level faults; Verify assigns success, failure, partial progress, or unresolved uncertainty. False acceptance, false rejection, calibration error, and verdict latency can be measured directly from annotated episodes. Claims about intervention benefit or recovery utility require matched runs, randomized verifier use, branching replay, or an explicit causal estimator. A credible evaluation reports accuracy together with evidence dependence, correlated failures, and the intervention window in which the verdict arrived.

3.6 Synthesis: From Capability Relations to MMEA Membership

PAPAV separates three levels of analysis. A *capability relation* specifies how a particular task responsibility is closed; a *system profile* records which of those relations are instantiated and observable; a *whole-system membership* rule determines whether the resulting configuration constitutes an MMEA. This distinction accommodates the overlap among multimodal agents, robotics, and generalist embodied agents. GEA, for example, spans games, user interfaces, and robotic embodiment [178]; its breadth illustrates why research-category overlap should be preserved while the operational MMEA criterion is stated separately rather than inferred from the breadth of a system's application domains.

The membership rule adopted here requires three conditions jointly. The task goal must persist across physical execution; each claimed capability return must have a documented route into the current task decision and be able to alter at least one decision under a specified state; and semantic commitments must be grounded in the deployed body and environment. Evidence is coded as documented, intervention-validated, or unreported, with intervention studies supporting the stronger causal interpretation. Membership is assigned only when all three conditions are supported. Missing evidence yields an indeterminate case, whereas an explicitly absent condition places the system outside the proposed class. Partial closures remain informative at the profile level, and end-to-end policies remain admissible when traces, ablations, or interventions expose the relevant relations. The propagation of uncertainty, execution state, and recovery cost across interfaces is treated separately as a dimension of system maturity and should be reported independently of membership.

Recent systems illustrate individual components of this account; validation of the taxonomy remains a separate empirical task. RoboAgent routes capability returns into a task scheduler, AGENTSAFE diagnoses failures across perception, planning, and execution, and EmbodiedEval extends interactive system-level evaluation [26, 198, 214]. Applying PAPAV reproducibly requires annotators to record, for every claimed capability, its input, return object, task-level responsibility, minimum observable evidence, and primary direct measure. Counterfactual diagnostics should be accompanied by their simulation, replay, pairing, or causal-identification assumptions. Agreement between independent coders on boundary systems then provides the appropriate test of whether these rules are sufficiently precise for comparative survey analysis.

4 Benchmarks & Evaluation

Having clarified the definitions of Multimodal Embodied Agents (MMEA), Multimodal Agents (MMA), and Robotic Systems (RS), and compared the three system types in terms of their PAPAV capability profiles, we provide a curated, structured comparison of how existing benchmarks cover and evaluate these capabilities.

It is important to distinguish between whether a capability is required to complete a task and whether it is explicitly evaluated by the benchmark. Complex tasks often require multiple capabilities, but some benchmarks report only task success rates or final artifact quality. Although such metrics reflect overall system performance, they cannot identify the exact capability responsible for a failure. Therefore, a capability being involved in task execution does not mean that it is explicitly evaluated.

Following the curation and classification procedure described below, we retained 62 benchmark entities, comprising 10 MMEA benchmarks, 22 MMA benchmarks, and 30 RS benchmarks. Table 2 summarizes their names, publication venues and years, task settings, and PAPAV capability-evaluation coverage. Unless otherwise stated, all counts and proportions reported in this section describe this curated corpus.

Benchmark Collection and Selection Candidate benchmarks were identified through iterative searches of Google Scholar and relevant conference and journal publications. Each included benchmark was manually verified against its primary paper, official publication page, or official project page. Inclusion decisions, bibliographic information, and PAPAV capability labels were determined from these primary sources. The corpus was last updated on 15 Aug 2026.

Candidate benchmarks were retained according to the following criteria:

1. Benchmarks should have a well-defined set of tasks or interaction environments.
2. Benchmarks should have publicly available and identifiable evaluation protocols and report at least one quantitative metric.
3. The tasks or metrics of the benchmarks should be related to at least one capability.
4. Benchmarks should provide sufficient publicly available information to determine whether each PAPAV capability is explicitly evaluated, implicitly involved, or not required by the task or evaluation protocol.

To ensure broad coverage within this curated comparison, we selected eligible benchmarks spanning diverse task settings, environment formats, capability objectives, and evaluation protocols.

Multiple publications describing the same benchmark release were consolidated and counted once, using the most complete version as the primary reference. A successor release was counted separately only when it introduced substantively different tasks, environments, datasets, or evaluation protocols. The MMEA, MMA, and RS categories were assigned mutually exclusively according to the benchmark's primary evaluation target and the definitions introduced above. Multimodal input alone did not cause a RS benchmark to be classified as MMEA.

Capability Labeling Protocol The capability-labeling protocol used in Table 2 follows the rules below. The unit of analysis was a benchmark–capability pair. Each of the 62 benchmark entities was assessed against each of the five PAPAV capabilities, yielding 310 coding units. Coding decisions were based on the primary benchmark paper and its official project page or official code documentation.

- **Explicit Evaluation (●)**: The benchmark provides capability-specific evidence through a dedicated task, metric, controlled comparison, or observable signal. This label does not imply that all failure causes can be uniquely attributed to that capability.
- **Implicit Evaluation (◐)**: The capability is required for task completion, but is reflected only through broader end-to-end performance without capability-specific evidence.
- **Not Evaluated (○)**: The capability is not required by the benchmark's task or evaluation protocol.

To avoid overestimating capability coverage, we further define the boundary rule for each capability:

1. **Perceive**: Perceive is Explicit when perceptual understanding, grounding, or the extraction of task-relevant information is assessed through a dedicated task, metric, or controlled comparison. It is Implicit when such information is necessary but reflected only through broader end-to-end performance. The presence of images, videos, or other multimodal inputs alone does not constitute Explicit Evaluation.
2. **Anticipate**: Anticipate is Explicit when predictions of future action consequences, changes in risk, or counterfactual states are required and scored. In interactive tasks, the fact that actions naturally change subsequent states is insufficient to establish Anticipate.
3. **Plan**: Plan is Explicit when task-level planning is directly assessed, such as goal decomposition, high-level skill organization, subtask coordination, or replanning. Path planning, motion planning, trajectory generation, and state-space search in robotics are not equivalent to task-level Plan.
4. **Act**: Act is Explicit when action execution, validity, or action-induced environmental effects are specifically assessed or scored. Offline action descriptions and future video generation do not constitute Explicit Act Evaluation. Requiring an agent to produce an action alone does not constitute Explicit Evaluation.
5. **Verify**: Verify is Explicit when the agent is required to determine success or failure from feedback. Verification performed by external test scripts, environment checkers, or automatic evaluators is not equivalent to the agent's Verify capability.

Table 2. Benchmark Overview and PAPAV Capability Coverage. The table contains the 62 retained benchmark entities (10 MMEA, 22 MMA, and 30 RS). Categories are mutually exclusive, and benchmark version and counting rules are reported in the collection protocol. Per., Ant., and Ver. denote Perceive, Anticipate, and Verify, respectively. ACL-F denotes Findings of ACL; CVPRW denotes CVPR Workshop; NeurIPS-DB denotes NeurIPS Datasets and Benchmarks.

Benchmark	Venue	Task	Per.	Ant.	Plan	Act	Ver.
Category: Multimodal Embodied Agent							
Subcategory: Simulation							
TEACh [150]	AAAI'22	Dialogue-guided tasks	◐	◐	◐	●	◐
RoCoBench [135]	ICRA'24	Collaborative coordination	○	◐	●	●	◐
PARTNR [19]	ICLR'25	Collaborative planning	●	◐	●	●	◐
EMBODIEDBENCH [203]	ICML'25	Embodied tasks	●	◐	●	●	◐
RoboCerebra [60]	NeurIPS'25	Long-horizon planning	◐	◐	●	●	●
Habitat-MAS [21]	ICLR'25	Multi-robot allocation	●	◐	●	●	◐
IS-Bench [124]	AAAI'26	Hazard mitigation	●	◐	◐	●	○

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Benchmark	Venue	Task	Per.	Ant.	Plan	Act	Ver.
ESI-Bench [67]	arXiv'26	Spatial intelligence	●	◐	●	●	○
Subcategory: Real							
PLanAR [54]	arXiv'26	Verified replanning	◐	◐	●	●	●
Subcategory: Hybrid							
CaP-Bench [42]	arXiv'26	Robot code refinement	●	◐	◐	●	◐
Category: Multimodal Agent							
Subcategory: Understanding							
MMMU-Pro [216]	ACL'25	Multimodal QA	●	○	○	○	○
VSI-Bench [202]	CVPR'25	Spatial reasoning	●	○	○	○	○
Video-MME-v2 [41]	arXiv'26	Video QA	●	○	○	○	○
Subcategory: Interaction							
WebArena [234]	ICLR'24	Web workflows	○	◐	◐	●	◐
GAIA [139]	ICLR'24	General AI Tasks	◐	○	◐	◐	○
VisualWebArena [94]	ACL'24	Visual web tasks	●	◐	◐	●	◐
WebVoyager [62]	ACL'24	Live web tasks	◐	◐	◐	●	◐
AndroidWorld [157]	ICLR'25	Android tasks	◐	◐	◐	●	◐
AgentStudio [231]	ICLR'25	Virtual-agent tasks	●	◐	◐	●	●
MMInA [183]	ACL-F'25	Multimodal web tasks	●	◐	◐	●	○
CRAB [199]	ACL-F'25	Cross-device workflows	◐	◐	◐	●	◐
iVISPAR [136]	EMNLP'25	Sliding-tile puzzles	●	◐	○	●	○
AgentVista [174]	arXiv'26	Visual agent tasks	●	○	◐	●	○
GameWorld [149]	arXiv'26	Browser-game tasks	◐	◐	◐	●	○
MMSearch-Plus [182]	ICLR'26	Multimodal retrieval	●	○	◐	●	○
OmniGAIA [106]	arXiv'26	Omni-modal QA	●	○	◐	●	○
OSWorld2.0 [215]	arXiv'26	Computer-use Task	◐	◐	◐	●	◐
Subcategory: Generation							
SWE-bench [84]	ICLR'24	Repository repair	◐	○	◐	●	○
WebGen-Bench [126]	NeurIPS-DB'25	Interactive website generation	◐	○	◐	●	○
PBench [145]	NVIDIA'25	Physical future prediction	◐	●	○	○	○
GameCraft-Bench [129]	arXiv'26	Playable game generation	◐	○	◐	●	○
WorldModelBench [101]	NeurIPS-DB'25	Future video generation	◐	●	○	○	○
Category: Robotic System							
Subcategory: Simulation							
Room-to-Room (R2R) [5]	CVPR'18	Instruction navigation	●	◐	○	●	◐
ALFRED [172]	CVPR'20	Household tasks	●	◐	◐	●	◐
RLBench [78]	RA-L'20	Manipulation learning	◐	◐	○	●	○
CALVIN [138]	RA-L'22	Instruction sequences	●	◐	○	●	○
BEHAVIOR-1K [100]	CoRL'22	Household activities	◐	◐	◐	●	◐
ManiSkill2 [49]	ICLR'23	Generalizable manipulation	◐	◐	○	●	○
LIBERO [114]	NeurIPS-DB'23	Lifelong manipulation	◐	◐	◐	●	○
Safety-Gymnasium [80]	NeurIPS'23	Safe control	◐	◐	○	●	○
GOAT-Bench [91]	CVPR'24	Multimodal navigation	●	◐	○	●	◐
RoboSuite [238]	arXiv'25	Policy training	◐	○	○	●	○
VLABench [224]	ICCV'25	Long-horizon manipulation	●	◐	◐	●	◐
Meta-World + [137]	NeurIPS-DB'25	Multi-skill learning	○	○	○	●	○
LIBERO-PRO [236]	arXiv'26	Robust manipulation	●	◐	◐	●	○
SafeManip [72]	arXiv'26	Safe manipulation	◐	◐	◐	●	○
Dream.exe [227]	arXiv'26	Future video execution	◐	●	○	◐	○

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Benchmark	Venue	Task	Per.	Ant.	Plan	Act	Ver.
RoboCasa365 [144]	ICLR'26	Kitchen activities	◐	◐	◐	●	○
RoboLab [206]	RSS'26	Robust manipulation	●	◐	◐	●	○
Subcategory: Real							
FurnitureBench [63]	RSS'23	Furniture assembly	◐	◐	◐	●	◐
FMB [128]	IJRR'24	Skill composition	◐	◐	◐	●	○
RoboArena [7]	CoRL'25	Policy comparison	◐	○	○	●	○
BEHAVIOR Robot Suite [83]	CoRL'25	Household manipulation	◐	◐	◐	●	◐
RoboChallenge [161]	Tech. Rep.'25	Policy evaluation	◐	○	○	●	○
VLA-REPLICA [71]	arXiv'26	Reproducible manipulation	◐	○	○	●	○
Subcategory: Hybrid							
HomeRobot [211]	CoRL'23	Mobile manipulation	●	◐	◐	●	◐
RoboTwin [142]	CVPR'25	Bimanual transfer	◐	○	○	●	○
RobotArena∞ [79]	ICLR'26	Reconstructed evaluation	●	○	○	●	○
RoboTwin 2.0 [22]	ICML'26	Bimanual policy generation	◐	○	○	●	○
RoboWM-Bench [81]	CVPRW'26	Future manipulation	◐	●	○	◐	○
NIABench [225]	IROS'26	Non-intrusive assistance	○	◐	○	●	○
RoboDojo [23]	arXiv'26	Sim-to-real evaluation	◐	○	○	●	○

Evaluation: ● Explicit ◐ Implicit ○ Not Evaluated.

4.1 Multimodal Embodied Agents

Because multimodal embodied tasks often involve several PAPAV capabilities simultaneously, we distinguish capability involvement from Explicit Evaluation, which requires capability-specific evidence but does not necessarily isolate every possible source of failure.

MMEA benchmarks are at the intersection of MMA and RS benchmarks, emphasizing how vision-language reasoning influences physical environment through actions. Table 2 reveals a relatively consistent capability distribution. All MMEA benchmarks explicitly evaluate Act capability, and most explicitly cover Perceive and Plan capabilities. In contrast, Anticipate capability is only implicitly evaluated across all benchmarks, and Verify capability is explicitly evaluated only by RoboCerebra [60] and PLanAR [54]. This indicates that current benchmarks and evaluation methods adequately cover the main "Perception – Planning – Action" chain, but diagnostics for predicting action consequences and verifying results are still insufficient. Furthermore, current benchmarks are primarily based on simulation environments, with limited real-world deployment and sim-to-real evaluation.

4.1.1 Embodied Perception-Action

IS-Bench [124], ESI-Bench [67], and CaP-Bench [42] all position multimodal perception within the closed-loop of embodied action, while emphasizing interaction safety, active spatial intelligence, and code-as-policy robotic control. IS-Bench requires agents to identify risks that dynamically emerge during household tasks. ESI-Bench requires agents to actively acquire task-relevant spatial information through movement and interaction. CaP-Bench evaluates whether agents can synthesize executable robot-control code from visual observations and language instructions, and iteratively refine it through environmental feedback.

These three benchmarks provide capability-specific evidence for Perceive and Act because their protocols assess task-relevant visual-state understanding and score action-induced environmental effects. Perceive and Act are therefore classified as Explicit. They evaluate Anticipate implicitly: risk avoidance, active observation, and reasoning about action consequences affect task completion, but none independently scores future-state prediction. Their treatment of Verify differs. IS-Bench and ESI-Bench do not evaluate agent-side Verify because their process checks and final-answer assessment do not require the agent to determine action success and respond by stopping, retrying, or recovering. CaP-Bench evaluates Verify implicitly because environmental feedback supports iterative code refinement, although verification itself is not independently scored.

The three benchmarks also differ in their evaluation of Plan. In IS-Bench and CaP-Bench, multi-step organization is reflected primarily in task outcomes, resulting in Implicit Plan evaluation. In contrast, ESI-Bench explicitly evaluates how agents select and sequence perception, locomotion, and manipulation behaviors to acquire task-relevant evidence. This comparison shows that closed-loop interaction can involve multiple PAPAV capabilities, but capability-level diagnosticity emerges only when intermediate reasoning, action organization, or feedback-driven decisions become identifiable evaluation targets.

4.1.2 Planning and Closed-loop

TEACh [150], EMBODIEDBENCH [203], RoboCerebra [60], and PLanAR [54] primarily examine Plan and Act capabilities in long-horizon embodied tasks. TEACh integrates language interaction, egocentric perception, and embodied execution in dialogue-guided household tasks. EMBODIEDBENCH spans household interaction, navigation, and manipulation, with dedicated settings for visual perception, spatial understanding, and long-horizon planning. RoboCerebra targets long-horizon robotic manipulation through a hierarchical architecture that combines a high-level vision-language planner with a low-level motion controller while incorporating memory and verification. PLanAR further integrates task planning, motion checking, and goal verification in real-world robotic environments to evaluate closed-loop replanning under perturbations.

TEACh explicitly evaluates Act, while Perceive, Anticipate, Plan, and Verify are evaluated implicitly through dialogue-guided, long-horizon task execution. EMBODIEDBENCH explicitly evaluates Perceive, Plan, and Act through capability-oriented subsets and interactive tasks, while Anticipate and Verify remain implicit in end-to-end execution and environmental feedback.

RoboCerebra and PLanAR provide more diagnostic closed-loop evaluation. Both explicitly evaluate Plan, Act, and Verify through mechanisms such as reflection, action checking, goal checking, and replanning. Perceive is implicitly evaluated because visual observations primarily support downstream planning and execution rather than serving as an independent perceptual objective. Similarly, Anticipate is mainly reflected through long-horizon decision-making and responses to perturbations. Overall, effective closed-loop evaluation requires an independent “Status Verification–Deviation Identification–Plan Revision” loop rather than merely longer task horizons.

4.1.3 Multi-agent Coordination

PARTNR [19], RoCoBench [135], and Habitat-MAS [21] extend embodied capabilities into multi-agent coordination. PARTNR targets human-robot collaboration in a household environment, with tasks involving spatial, temporal, and heterogeneous capability constraints. RoCoBench requires multiple robots to form collaborative strategies, subtask assignments, and execution plans. Habitat-MAS focuses on heterogeneous robots with different embodied and motion capabilities, evaluating task allocation and coordinated execution in a multi-story household environment.

All three benchmarks explicitly evaluate Plan and Act through capability-specific coordination and execution evidence. PARTNR and Habitat-MAS also provide Explicit Perceive evidence through tasks or analyses that assess environmental-state interpretation and the grounding of heterogeneous agent capabilities. In contrast, RoCoBench largely provides textual or structured state representations to the planning module and therefore does not evaluate multimodal perception.

Multi-agent coordination also introduces implicit requirements for Anticipate. Dependencies, collision constraints, and heterogeneous capabilities require agents to reason about other agents' actions as well as the consequences, but such anticipatory reasoning is rarely evaluated as an independent target. Similarly, Verify is mainly reflected through environmental feedback, task progress, and recovery from coordination failures. External collision checking or simulator-based validation, however, should not be considered as an agent's own ability to verify collaborative outcomes.

Overall, recent MMEA benchmarks increasingly move beyond final task outcomes by evaluating intermediate behaviors such as active perception, planning, verifying, replanning, coordination, and recovery. Across these benchmarks, Perceive, Plan and Act are generally the most explicitly evaluated capabilities. In contrast, Anticipate and Verify are still more often embedded in task execution. This suggests that the key remaining challenge is not broader task coverage, but finer capability-level observability that can reveal where and why an embodied agent succeeds or fails.

4.2 Multimodal Agents

MMA benchmarks are evolving from capability-specific evaluation toward interactive and closed-loop settings. Within the MMA category, multimodal-understanding benchmarks primarily assess Perceive [41, 202, 216], whereas benchmarks for web, computer-use, mobile-platform, and game interaction explicitly evaluate Act while also implicitly covering Plan and Anticipate [149, 157, 215, 234]. Artifact-generation benchmarks further extend Act to the production of executable artifacts [84, 126, 129], while world model benchmarks evaluate Anticipate through future-state prediction [101, 145]. Overall, MMA benchmarks exhibit broad capability coverage, but the evaluation of Verify remains comparatively limited. Although several benchmarks provide post-action feedback [136, 149, 182, 183], they rarely require agents to explicitly verify whether an action has been successfully completed and subsequently decide whether to stop, retry, or recover based on the verification result. Accordingly, we distinguish between state information available to the agent and privileged state or rules used only by the benchmark evaluator. State checking by an evaluator alone does not constitute evidence of the agent's Verify capability.

4.2.1 Multimodal Understanding

MMMU-Pro [216], Video-MME-v2 [41], and VSI-Bench [202] target interdisciplinary multimodal question answering, video understanding, and visual-spatial reasoning, respectively. MMMU-Pro reduces the influence of linguistic priors on results by filtering out questions that can be answered from text information alone. Video-MME-v2 evaluates video understanding through visual information aggregation, temporal dynamic modeling, and cross-temporal reasoning. VSI-Bench evaluates spatial relations, distance, orientation, and scene memory based on egocentric indoor video.

These three benchmarks explicitly evaluate Perceive, but do not evaluate the other four capabilities. Although the tasks involve knowledge-based, temporal, or spatial reasoning, models operate on fixed multimodal inputs and only need to produce answers. These benchmarks primarily provide capability-specific evaluation instead of complete agent-loop evaluation.

4.2.2 Web Browsing, Computer Use, and Mobile Interaction

WebArena [234], VisualWebArena [94], WebVoyager [62], MMInA [183], and MMSearch-Plus [182] integrate agent evaluation with web browsing and search. WebArena uses structured webpage representations and therefore does not evaluate perception of the original visual webpage. VisualWebArena, MMInA, and MMSearch-Plus provide Explicit Perceive evidence through visual grounding or multi-modal evidence-acquisition tasks. WebVoyager requires visual webpage observations, but perceptual understanding is reflected only through end-to-end task success and is therefore classified as Implicit.

These benchmarks explicitly evaluate Act through multi-step interaction and provide implicit coverage of Plan. In WebArena, VisualWebArena, WebVoyager, and MMInA, page transitions, cross-site dependencies, and potentially irreversible operations require agents to account for the consequences of their actions, thereby implicitly covering Anticipate. MMSearch-Plus primarily focuses on evidence retrieval and integration and does not directly require future-state prediction.

WebArena, VisualWebArena, and WebVoyager also require agents to inspect the webpage state after each action and decide whether to continue, correct, or terminate the task, providing implicit evaluation of Verify. MMInA's hop-by-hop success rate is computed by the evaluator and therefore does not represent agent-side verification. MMSearch-Plus's Evidence Accuracy evaluates the quality of evidence and is not equivalent to action result verification. Therefore, these two do not cover Verify.

AndroidWorld [157], OSWorld 2.0 [215], and AgentStudio [231] extend interaction to mobile devices, desktop systems, and general virtual environments. AndroidWorld and OSWorld 2.0 explicitly evaluate Act while Perceive, Anticipate, and Plan are reflected implicitly through long-horizon execution with dynamic interfaces. Both also require agents to infer task completion from intermediate states and decide on subsequent steps, thereby providing Implicit evaluation of Verify.

AgentStudio offers more fine-grained capability diagnostics. Its GroundUI, IDMBench, and CriticBench evaluate interface localization, video-based action learning, and task success verification, respectively, thereby explicitly evaluate Perceive, Act, and Verify, while Anticipate and Plan are still implicitly covered. In particular, CriticBench treats success judgment itself as an evaluation target, making Verify an explicitly evaluated capability.

4.2.3 General and Cross-environment Assistants

GAIA [139], CRAB [199], AgentVista [174], and OmniGAIA [106] extend agent evaluation to cross-tool, cross-device, and cross-modal tasks. GAIA evaluates document understanding, web search, reasoning, and tool use through real-world questions with deterministic answers. CRAB supports cross-environment workflows spanning desktop and mobile devices, while AgentVista emphasizes visual evidence-driven hybrid tool use. OmniGAIA further extends evaluation to multimodal reasoning across images, video, and audio.

In GAIA, multimodal understanding, tool invocation, and multi-step reasoning are primarily evaluated through final answer accuracy. Therefore, Perceive, Plan, and Act are only evaluated implicitly. Its tasks generally do not require agents to anticipate the consequences of their own actions or verify intermediate outcomes. In contrast, CRAB provides Explicit Act evidence through graph-based checks of action-induced task progress. Perceive is Implicit because visual-state interpretation affects task completion but is not assessed through capability-specific evidence. Cross-device dependencies and multi-stage execution also provide Implicit evaluation of Anticipate, Plan, and Verify.

AgentVista and OmniGAIA explicitly evaluate Perceive and Act, while Plan is implicitly reflected in multi-step tool use. Their tools primarily support information acquisition rather than requiring anticipation of action-conditioned future states. Therefore Anticipate is Not Evaluated. Their final answers and supporting evidence are assessed externally, so Verify is also Not Evaluated.

4.2.4 Interactive Games and Spatially Intelligent Agents

GameWorld [149] and iVISPAR [136] place visuospatial understanding within interactive environments. GameWorld covers multiple browser-based games, requiring agents to continuously process visual feedback and execute keyboard, mouse, or semantic actions. iVISPAR uses sliding puzzles to compare spatial reasoning and sequential manipulation across 2D visual, 3D visual, and textual state representations.

Both benchmarks provide Explicit Act evidence through scored action-induced state changes. They provide Implicit Anticipate evidence because agents must account for how their actions affect subsequent states. iVISPAR provides Explicit Perceive evidence through its controlled comparison of 2D visual, 3D visual, and textual state representations. In GameWorld, visual processing is required but is reflected only through state-verifiable task success and progress, therefore Perceive is only Implicit. GameWorld also provides Implicit Plan evaluation through long-horizon objectives and multi-stage behavior organization, whereas iVISPAR's state-space search and local move sequences do not constitute task-level Plan.

Neither benchmark evaluates Verify capability. In iVISPAR, observing the board state after a move is a typical observe-control loop, without requiring independent success judgments or error recovery.GameWorld's state-verifiable evaluation shows that the evaluator can reliably check the game state, but it does not require the agent to verify the outcomes of its own actions.

4.2.5 Artifact Generation

SWE-bench [84], WebGen-Bench [126], and GameCraft-Bench [129] evaluate software issue resolution, interactive website generation, and playable game generation, respectively. In these tasks, Act extends to generating artifacts that can be compiled, run, or interactively tested.

All three benchmarks require the agent to understand task descriptions, code states, and artifact requirements. Perceive is evaluated implicitly because this understanding is not assessed through an independent objective or metric. Task-level organization also affects the correctness and completeness of the resulting patch, website, or game, but the plan itself is not independently assessed. Therefore Plan is Implicit in all three benchmarks.

None of the three evaluates Anticipate or Verify. Unit tests, webpage tests, and playability checks evaluate artifact quality from external evaluators, but do not require agents themselves to predict the consequences of generated changes or to actively test outputs, identify failures, and iteratively revise them. Such external validation should therefore be distinguished from agent-side anticipation and verification.

4.2.6 World Model and Physical Prediction

WorldModelBench [101] and PBench [145] evaluate future video generation conditioned on initial visual states and textual inputs, with a focus on temporal consistency, physical plausibility, and condition compliance. Both explicitly evaluate Anticipate, as predicting future states and environmental dynamics constitutes the primary task output. Although understanding the initial scene is necessary for accurate prediction, perception is not assessed through independent objectives or metrics, and therefore Perceive is evaluated only implicitly.

Future video generation does not constitute action in the environment, and neither benchmark evaluates Act. They also do not require models to formulate task-level plans based on predicted futures or to verify and revise predictions through subsequent environmental feedback, leaving Plan and Verify Not Evaluated. These benchmarks provide targeted diagnosis of future-state prediction, but do not yet integrate prediction with subsequent planning, action, and verification into a complete agent loop.

Overall, MMA benchmarks have expanded from capability-specific evaluation toward increasingly interactive and multi-step settings, resulting in broad PAPAV coverage. Perceive and Act are the most explicitly evaluated capabilities across multimodal understanding and interactive environments. Plan is commonly embedded in multi-step tasks but is less often explicitly evaluated as an independent capability. Anticipate is usually implicit in action-dependent tasks and becomes explicit mainly in future-state prediction benchmarks. Verify remains the least evaluated capability, as most benchmarks rely on environmental feedback or external outcome checking instead of agent-side verification and recovery.

4.3 Robotic Systems

The measurement focus of RS benchmarks is relatively concentrated. Of the 30 benchmarks in Table 2, 28 provide Explicit Act evidence through scored task outcomes, return, path efficiency, or stage completion resulting from robot actions. These metrics make action effects observable, although aggregate scores do not uniquely distinguish execution failures from upstream perception or planning failures. None provides Explicit Evaluation of task-level Plan or agent-side Verify. Path planning, motion planning, trajectory optimization, and low-level control are treated as execution-level optimization rather than task-level Plan unless the benchmark separately evaluates goal decomposition, subgoal organization, skill sequencing, or replanning. Existing protocols can thus describe what a robot accomplished more readily than the decisions that preceded execution or the checks that followed it. This asymmetry affects failure analysis: the same low success rate may result from an unsuitable goal interpretation, a poor skill choice, or inaccurate control. We treat simulation, physical deployment, and hybrid testing as evaluation settings, since the same navigation or manipulation problem may appear in more than one of them.

4.3.1 Navigation and Exploration

R2R evaluates instruction following from egocentric vision and combines endpoint success with path efficiency [5]. GOAT-Bench replaces a single route with a continuing sequence of category, language, and image goals, making spatial memory part of the task [91]. Earlier exploration can support later goals, so the episode structure tests information reuse in addition to immediate instruction following. Both provide capability-specific evidence for visual grounding and navigation, so Perceive and Act are Explicit. Their aggregate scores still mix several failure sources: an agent may miss the target, localize itself incorrectly, lose information from earlier exploration, or execute an unsuitable movement.

State prediction and progress checking are present in a successful trajectory, yet neither receives a separate score. Anticipate and Verify are consequently inferred from behavior and marked Implicit. We also keep path search outside task-level Plan. This boundary distinguishes navigation decisions from the decomposition, subgoal organization, and replanning studied in other benchmark families.

The contrast between R2R and GOAT-Bench also shows why episode design matters. A similar navigation score can reflect single-route instruction following in one benchmark and accumulated map use in the other. Memory-specific errors and goal-level trajectories would make this distinction easier to inspect.

4.3.2 Skill Learning and Policy Generalization

RLBench, ManiSkill2, RoboSuite, and Meta-World+ establish common manipulation tasks, robot models, controllers, or skill collections for comparing policy learning and basic generalization [49, 78, 137, 238]. Their central measurements are success and return, which make Act readily observable but combine the causes of failure into a policy-level result.

Other suites sharpen the generalization question. LIBERO separates the acquisition of new tasks from retention of earlier skills, and LIBERO-PRO adds visual and task perturbations [114, 236]. CALVIN evaluates language-conditioned skill sequences, while RoboCasa365 scales such sequences to varied kitchen activities [138, 144]. RoboLab introduces controlled changes to tasks and environments [206]; FMB moves individual and composed skills onto a reproducible physical platform [128].

These protocols probe different notions of generalization. Continual learning separates acquisition from retention, perturbation tests measure local stability, and sequential tasks examine whether skills survive changes in order and context. FMB also introduces calibration, contact, and reset variation. Each score should be interpreted within its own protocol.

Cross-suite comparisons must preserve the generalization axis being tested. A policy that withstands visual perturbations may still forget earlier tasks or fail when familiar skills are composed into a longer sequence. Aggregate success does not distinguish these behaviors.

CALVIN, LIBERO-PRO, and RoboLab deliberately manipulate grounding or visual conditions, providing Explicit Perceive evidence. In most remaining suites, Perceive is embedded in end-to-end success and is therefore Implicit. Meta-World+ is the exception because its protocol uses structured states and does not evaluate Perceive. Plan is Implicit in LIBERO, LIBERO-PRO, RoboCasa365, RoboLab, and FMB, although none independently evaluates the plan itself. In RLBench, ManiSkill2, RoboSuite, Meta-World+, and CALVIN, the evaluated protocol does not require independently identifiable task-level planning, so Plan is Not Evaluated. Verify is Not Evaluated throughout this group because benchmark-side completion checking does not demonstrate agent-side failure recognition or recovery.

4.3.3 Long-horizon and Household Manipulation

ALFRED and VLABench connect language, navigation, and manipulation in multi-stage tasks, where a valid action can still fail when performed at the wrong point in the sequence [172, 224]. BEHAVIOR-1K and HomeRobot expand the task space to object search, mobile manipulation, and everyday activities [100, 211]. FurnitureBench and the BEHAVIOR Robot Suite place long-horizon assembly and household tasks on physical platforms, adding contact errors, hardware tolerances, and reset variation [63, 83].

Goal conditions and phase-level progress give a more informative account than terminal success alone. They reveal where an episode stopped and distinguish an early breakdown from near completion. Even so, a single unfinished task may contain semantic, perceptual, navigation, and manipulation errors. Progress is also computed by an external evaluator. It describes how far the robot moved through the prescribed task, without revealing how the robot organized the steps or whether it recognized a failed intermediate state. Plan and Verify thus remain Implicit.

Phase reporting is especially useful when full completion is rare, as in physical assembly. It preserves partial progress, although the chosen phase definitions still limit how precisely a failure can be attributed.

Subgoal-level traces could sharpen this signal by pairing each failed phase with its intended preconditions and executed actions. Most current protocols stop at stage completion and leave the underlying capability error unresolved.

4.3.4 Safety-constrained Control and Human Assistance

Safety-Gymnasium reports task reward alongside constraint cost, allowing an unsafe high-return policy to be distinguished from a safe one [80]. SafeManip checks prohibited states and temporal-ordering violations during manipulation [72]. NIABench evaluates the utility and timing of assistance, treating inaction as a meaningful policy choice when intervention would be unnecessary or disruptive [225].

These protocols broaden Act beyond simple task completion: the consequences of an action must also be safe and appropriate. Risk may shape the selected behavior, but the policy's forecast is not scored, leaving Anticipate Implicit. SafeManip's ordered constraints provide indirect evidence of task-level organization, so Plan is Implicit. Safety-Gymnasium and NIABench do not require task-level Plan under our coding boundary. Because constraint monitoring and intervention scoring remain external to the evaluated agent, none of the three evaluates agent-side Verify.

In assistance tasks, a technically successful intervention may still reduce utility by interrupting a person. NIABench measures this trade-off; Safety-Gymnasium and SafeManip emphasize cumulative violations and temporal constraints.

4.3.5 Real-world and Sim-to-real Evaluation

RoboArena distributes pairwise physical-policy comparisons across participating sites, reducing dependence on the task setup of a single laboratory [7]. VLA-REPLICA specifies a lower-cost hardware configuration for repeatable in- and out-of-distribution tests [71], whereas RoboChallenge organizes centralized evaluation at larger scale [161]. RoboTwin and RoboTwin 2.0 use digital twins to expand scene and bimanual variation before deployment [22, 142]. RobotArena∞ asks whether policy rankings in reconstructed simulations agree with physical trials [79]; RoboDojo connects simulation with standardized remote robot experiments [23].

The shared concern is deployment validity across sites, hardware, and sim-to-real conditions. Camera placement, control frequency, reset procedures, and operator involvement can all change the result. Ranking stability, repeated-trial variance, and protocol disclosure are consequently central to interpreting success rates. Every benchmark in this group explicitly evaluates Act. However, task generation, state checking, and ranking computation belong to the evaluation infrastructure and do not constitute policy-level evidence for Plan or Verify. Perceive becomes Explicit only when visual variation is itself controlled and compared.

Agreement between simulated and physical policy rankings can be more informative than absolute simulated success. A simulator may overestimate every policy yet preserve their relative order. A high score under one generated distribution also says little about cross-site robustness when hardware, viewpoints, and reset conditions change. These benchmarks place such variation inside the evaluation protocol.

Site-level results are particularly useful here. A single pooled score can conceal a policy that performs consistently in one laboratory and degrades sharply under another site's hardware or reset procedure.

4.3.6 Robot World Models and Physical Prediction

Dream.exe evaluates generated manipulation futures through video quality, trajectory consistency, and downstream execution [227]. RoboWM-Bench likewise asks whether predicted robot futures are physically plausible and useful for manipulation [81]. Both make the future state an evaluation object, giving Explicit Anticipate evidence. The initial image serves only as an input condition, so Perceive remains Implicit; downstream execution tests the utility of the prediction and provides indirect evidence for Act.

The current protocols largely end after establishing that a prediction is useful. A fuller decision loop would require the robot to compare alternative futures, select an action, observe the resulting state, and update its next prediction. Since the evaluated world models do not complete this sequence, task-level Plan and Verify remain Not Evaluated.

Downstream execution shows that a predicted future retains task-relevant information; control remains the responsibility of a separate policy. Recording which prediction a controller selected, and how the observed transition changed later predictions, would expose the missing link between Anticipate and closed-loop behavior.

4.4 Cross-domain Comparison

Figure 3 summarizes the 62 benchmarks retained in our curated corpus in two views. The left Sankey diagram traces 49 outcome-based benchmarks (24 outcome-only and 25 outcome-augmented) and 13 process-diagnostic benchmarks to MMA, MMEA, and RS; flow width encodes the number of benchmarks. The right matrix arranges system families by row and PAPAV capabilities by column. Each pie reports the within-cell proportions of Explicit (green), Implicit (yellow), and Not Evaluated (red) labels, allowing comparison across differently sized families. Explicit indicates capability-specific evidence from a dedicated task, metric, controlled comparison, or observable signal; Implicit indicates that the capability is required but reflected only through broader end-to-end performance; and Not Evaluated indicates that the capability is not required by the protocol.

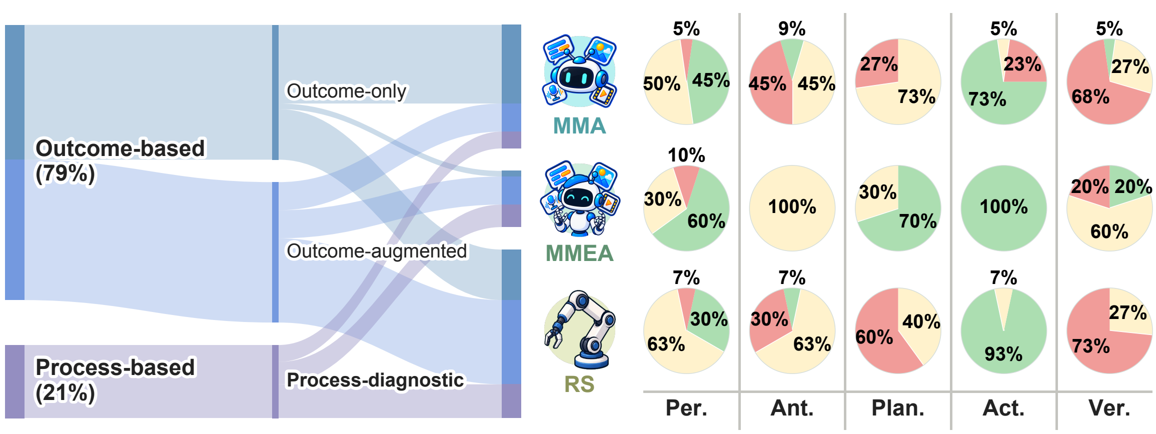

Figure 3. Analysis of benchmark metrics and cross-domain PAPAV coverage. Left: Sankey diagram tracing 62 benchmarks by evaluation basis, metric type, and system category. Right: PAPAV coverage matrices for MMA, MMEA, and RS.

4.4.1 Overall PAPAV Coverage

Because the retained categories have unequal sizes (10 MMEA, 22 MMA, and 30 RS benchmarks), cross-family percentages should be interpreted descriptively and together with their underlying counts. They characterize the current curated corpus and do not support statistical inference about the complete benchmark literature.

Act is the strongest common feature. It is Explicit in 16/22 MMA benchmarks, 28/30 RS benchmarks, and 10/10 MMEA benchmarks. Web clicks, software operations, and robot controls produce observable and scored changes in their environments. Perceive is Explicit in 10/22 MMA benchmarks, 9/30 RS benchmarks, and 6/10 MMEA benchmarks. In MMA benchmarks, visual input alone does not establish Explicit Perceive evidence. The Explicit cases instead include capability-specific multimodal-understanding, grounding, or evidence-acquisition tasks. Robotic benchmarks more often embed vision inside end-to-end policies, where perceptual and control errors share the same outcome score.

Plan produces the clearest separation between families. Sixteen MMA and twelve robotic benchmarks evaluate it implicitly, yet neither family contains an Explicit task-level Plan label. Seven MMEA benchmarks explicitly evaluate decomposition, coordination, action sequencing, or replanning. This pattern follows from protocol design: planning errors become identifiable only when the plan or an attributable planning signal is evaluated. It does not imply that one system family possesses stronger planning ability.

For Plan, 7/10 (70%) MMEA benchmarks are Explicit and 3/10 (30%) are Implicit. Among MMA benchmarks, 16/22 (73%) are Implicit and 6/22 (27%) are Not Evaluated. Among RS benchmarks, 12/30 (40%) are Implicit and 18/30 (60%) are Not Evaluated. These distributions characterize how Plan is evaluated within the curated corpus rather than the planning ability of the corresponding system families.

Evidence for Anticipate and Verify is much scarcer. Within the 62-benchmark curated corpus, only four benchmarks explicitly evaluate Anticipate and three explicitly evaluate Verify. A state change does not reveal whether the agent forecast it, and an automatic success function does not show whether the agent understood the outcome. Anticipate requires a scored prediction of future states or action consequences; in interactive settings, stronger evidence is provided when that prediction affects action selection. Verify requires a consequential response to feedback, such as stopping, retrying, repairing an output, or replanning.

4.4.2 Diagnosticity and Capability Bottlenecks

Broad capability coverage does not guarantee useful error diagnosis. A complex end-to-end task may involve perception, planning, and action while reporting only one success score. WorldModelBench and PBench use a narrower scope to isolate physical future prediction; AgentStudio's CriticBench similarly isolates success detection [101, 145, 231]. RoboCerebra records hierarchical decisions and checking signals, and PLanAR scores replanning after execution deviations [54, 60]. These process measures help separate a poor plan, a missed failure, and faulty execution.

The same distinction clarifies the current bottlenecks. Dream.exe and RoboWM-Bench score physically useful predictions, but stop before an agent compares alternatives, acts on one, and updates its forecast from feedback [81, 227]. Unit tests, simulator checkers, and safety monitors verify outcomes for the benchmark; agent-side Verify becomes observable only when that judgment changes subsequent behavior. Explicit Plan likewise needs scored decomposition, subgoal order, constraints, coordination, or replanning. Long trajectories and motion planning alone cannot provide that evidence. Controlled visual variation, localized grounding scores, and action-level traces remain valuable even for the better-covered Perceive and Act stages, since terminal success can hide errors in either one.

4.4.3 Convergence across Domains

Across the interactive benchmarks retained in this corpus, similar evaluation settings appear across domains. VisualWebArena and OSWorld 2.0 move MMA evaluation toward persistent interaction with digital states [94, 215]. RoboCasa365 adds language and compositional tasks to robotics, while RoboTwin expands environment generation through digital twins [142, 144]. MMEA benchmarks combine these directions: EMBODIEDBENCH spans several embodied settings, and PARTNR makes collaborative planning and heterogeneous constraints part of the scored interaction [19, 203].

As tasks become more integrated, a single success score explains less about the source of failure. End-to-end performance should be accompanied by selected process evidence—predictions, subgoal progress, failure detection, and recovery decisions—with agent behavior kept distinct from external evaluation. Such reporting would preserve system-level comparison while locating deficiencies within the PAPAV profile. No benchmark needs to evaualte every PAPAV capability, the essential requirement is a clear account of which stages are explcit evaluate and which are inferred from task success.

Overall, the curated benchmarks expose multi-capability coupling across digital and physical domains, while their evaluation remains centered primarily on task outcomes. This observation is a synthesis of the retained literature rather than evidence from a longitudinal analysis of field-wide trends.

4.5 Evaluation Metrics and Reproducibility

This section analyzes existing benchmarks from the perspectives of evaluation metrics and reproducibility. Evaluation metrics determine whether overall performance can be decomposed to identify specific capability deficiencies. Reproducibility affects reliability and comparability because benchmarks vary substantially in task space, interaction interface, and execution environment. Consequently, raw scores are generally meaningful within the same benchmark but are not directly comparable across benchmarks.

4.5.1 Outcome- and Process-based Evaluation

The evaluation metrics used by existing benchmarks can be broadly categorized into two groups: outcome-based and process-based evaluation. Outcome-based evaluation focuses primarily on final task results and can be further divided into two types: outcome-only and outcome-augmented. Outcome-only evaluation considers only the final result, commonly using metrics such as answer accuracy [139], task success rate [138, 234], test pass rate [84], and artifact quality [126, 129]. Outcome-augmented evaluation additionally incorporates auxiliary information about progress, efficiency, or cost while retaining task completion as the primary evaluation target [5, 91, 172]. In contrast, process-based evaluation corresponds to process-diagnostic evaluation, which explicitly assesses intermediate decisions, states, actions, or failure types, such as sub-goal progress, intermediate-state assessment, failure detection, and re-planning [54, 124, 199], thereby enabling more precise error localization.

For corpus coding, these categories were applied hierarchically. Reporting multiple metrics, such as success, completion time, and collision counts, does not by itself make a benchmark process-diagnostic. A benchmark was classified as process-diagnostic if it included at least one dedicated metric, annotation, intermediate-state assessment, or failure category that could attribute performance to a particular decision, stage, or capability. Otherwise, it was classified as outcome-augmented if terminal task performance remained the primary target but was accompanied by non-attributable measures of progress, efficiency, cost, or constraint violations. Benchmarks reporting only terminal success, answer correctness, test passing, return, or final-artifact quality were classified as outcome-only.

As shown in the left panel of Figure 3, within the 62-benchmark curated corpus, 24 benchmarks are outcome-only and 25 are outcome-augmented. Thus, 49 of 62 benchmarks (79%) primarily rely on outcome-based evaluation, while 13 of 62 (21%) provide process-diagnostic metrics. Within each retained system category, process-diagnostic evaluation appears in 4 of 10 MMEA benchmarks (40%), 6 of 30 RS benchmarks (20%), and 3 of 22 MMA benchmarks (14%). These values are descriptive statistics for the curated corpus and should not be interpreted as prevalence estimates for the complete benchmark literature. Within this sample, process-level evaluation is more visible in MMEA, while outcome-based evaluation remains dominant overall.

Outcome-based metrics are intuitive and easy to interpret, but provide limited support for distinguishing failures across PAPAV capabilities. Process-diagnostic evaluation enables more precise capability attribution, but requires finer-grained state annotations, trajectory logging, and evaluator design. Future benchmarks should therefore further integrate final task outcomes with key process-level metrics, enabling more informative error diagnosis while preserving the interpretability and comparability of overall performance.

4.5.2 Reproducibility

Benchmark reproducibility depends on whether task data, execution environments, and evaluators can be reproduced consistently. Static question answering and offline generation tasks typically use fixed inputs and deterministic scoring, making them relatively straightforward to reproduce. Self-hosted websites, operating systems, and simulated robotic environments provide greater control over initial states and task versions, but remain sensitive to changes in browsers, software dependencies, simulator versions, and external model interfaces. Real-robot evaluation introduces additional sources of variation, including hardware differences, camera calibration, control frequency, scene configuration, and human intervention, making cross-laboratory reproduction substantially more challenging.

Existing benchmarks improve evaluation stability through procedural success checkers, environment snapshots, simulator reconstruction, and standardized hardware configurations [71, 157, 234]. RoboArena [7] and RobotArena∞ [79] further explore distributed real-world evaluation and real-world scenario reconstruction, although their results may still vary with task selection, execution conditions, and evaluator implementation. Thus, releasing data and code alone is insufficient to guarantee complete reproducibility.

To improve reproducibility, benchmarks should disclose task instances, environment versions, evaluator implementations, and raw execution trajectories, while reporting model versions, prompt configurations, observation and action interfaces, step or time budgets, random seeds, numbers of trials, and hardware settings. For benchmarks relying on human or model-based evaluators, evaluation criteria, consistency checks, and evaluator versions should also be documented. Standardizing such experimental metadata is more important for establishing reliable and reproducible evaluation than directly comparing raw success rates across heterogeneous benchmarks.

For the reproducibility of this survey analysis, the benchmark corpus was frozen at the stated update cutoff. Table 2 reports the 62 retained benchmark entities together with their category assignments and PAPAV labels, allowing the reported counts and proportions to be recomputed. The collection, version-consolidation, category-assignment, and labeling rules are specified earlier in this section.

4.6 Discussion and Future Benchmark Design

The preceding analysis shows that the main limitation of current benchmarks is no longer insufficient task complexity, but insufficient evaluation granularity. Existing benchmarks increasingly involve multimodal, long-horizon, and interactive tasks across digital and embodied environments [19, 150, 215, 234], while their evaluation often remains dominated by aggregate task outcomes. Future benchmark design should therefore focus on making capability contributions and failure sources more identifiable rather than simply expanding task scope.

4.6.1 From Capability Coverage to Evaluation Identifiability

Broad PAPAV involvement does not necessarily provide informative diagnosis. A benchmark may require perception, planning, and action simultaneously, but a single success score cannot distinguish which capability caused failure. In contrast, benchmarks such as RoboCerebra and PLanAR expose intermediate checking, replanning, or recovery signals that provide more direct evidence of where failures occur [54, 60]. Future benchmarks should preserve end-to-end task performance while incorporating a set of attributable intermediate signals, such as grounding correctness, failure recognition, and recovery effectiveness. Controlled perturbations and intermediate checkpoints can further isolate capability-specific weaknesses without requiring every benchmark to explicitly evaluate the complete PAPAV loop.

4.6.2 Evaluating Capability Transitions and Adaptation

The current coverage also reveals that Perceive and Act are relatively observable, whereas Anticipate, Plan, and Verify are more often embedded within task execution. WorldModelBench and PBench, for example, explicitly evaluate future-state prediction, but stop before connecting prediction to subsequent action and verification [101, 145]. Future evaluation should therefore pay greater attention to transitions between capabilities. Anticipation becomes meaningful when predicted consequences affect action selection, planning when feedback leads to replanning, and verification when detected failures trigger stopping or recovery. Evaluating these transitions can reveal whether agents merely complete predefined interaction sequences or can genuinely adapt to changing environments.

4.6.3 Balancing Realism and Reproducibility

Increasing environmental realism improves deployment validity but also introduces variability from software, simulators, hardware, calibration, and execution conditions. Recent real-world and sim-to-real benchmarks such as RoboArena and RobotArena∞ increasingly expose these sources of variation across sites and environments [7, 79]. Future benchmarks should therefore combine controlled and reproducible core settings with progressively more diverse perturbation or real-world evaluation. Standardized reporting of task instances, environment versions, trajectories, evaluator implementations, and execution configurations is essential for interpreting results reliably.

Overall, future benchmarks should prioritize observable, attributable, and reproducible interaction. The objective is not to maximize PAPAV coverage, but to ensure that the capabilities central to a task can be meaningfully evaluated and that failures can be traced to specific stages or transitions of the interaction process.

5 Open Challenges

The preceding sections show that multimodal embodied agents increasingly possess the individual capabilities required to perceive, predict, plan, act, and evaluate. Yet stronger components do not by themselves resolve several architectural choices that arise when these capabilities must operate together in a closed loop. In particular, an embodied agent must decide what information should persist beyond the current observation, how long it should commit to an action before reconsidering its decision, how execution should be verified, what aspects of the future are worth predicting, and how uncertainty should change its behavior. We highlight five challenges that we believe are central to the next generation of multimodal embodied agents.

5.1 OC1. Persistent State and Memory

Core question. *How should an embodied agent maintain, update, and forget a persistent belief about the world under partial observability?*

A single observation rarely contains all information required for embodied decision making. Objects may become occluded, properties may only be observed from particular viewpoints, and the consequences of previous actions may remain relevant long after the corresponding observation has disappeared. Classical robotic systems address part of this problem through persistent geometric and semantic representations such as SLAM and semantic maps [16, 162]. Recent embodied systems extend this idea toward open-vocabulary scene representations and continuously updated spatial beliefs [4, 50, 125, 134]. The open problem is no longer simply whether an agent should have memory, but *what form that memory should take and how it should evolve over time*.

Explicit structured belief. One direction is to explicitly represent entities, relations, physical states, and possibly uncertainty in object-centric maps, scene graphs, or other structured world models [50, 134, 197]. Such representations are interpretable and directly queryable: an agent can, for example, record that a cup was previously on a table even after the cup becomes occluded. They also make updates and corrections conceptually explicit. However, persistent structured memories must determine which entities deserve to be retained, how contradictory observations should be reconciled, and when stale information should be removed. Without such mechanisms, memory can grow indefinitely while accumulating outdated or mutually inconsistent state.

Latent recurrent state. An alternative is to encode history implicitly inside a recurrent or latent belief state. Rather than storing a human-readable description of every relevant object and event, the agent learns a compact representation that summarizes information required for future decisions [8, 213]. This can provide a differentiable and computationally compact interface to the policy, but makes it more difficult to inspect what has been retained, explicitly correct erroneous beliefs, or guarantee that information important over long time horizons has not been forgotten.

Retrieval-based episodic memory. A third direction is to preserve past observations, interactions, or trajectories and retrieve them when they become relevant [123, 230]. This avoids requiring the agent to compress its entire history into a fixed state and allows detailed evidence from the past to be recovered on demand. However, it shifts the problem toward deciding *what to store, when to retrieve it, and how retrieved experience should modify the current belief.*

These approaches need not be mutually exclusive. A practical embodied agent may combine a compact latent state for continuous control, a structured world model for persistent entities and relations, and episodic storage for detailed historical experience. The unresolved question is therefore not simply how to increase memory capacity, but how to construct a persistent belief that is simultaneously *compact, editable, queryable, uncertainty-aware, and capable of forgetting*.

5.2 OC2. Planning and Execution Granularity

Core question. *At what semantic and temporal granularity should an embodied agent plan and execute actions?*

Planning granularity determines how long an agent commits to a decision before new observations are allowed to change its behavior. At one extreme, a model can translate an instruction directly into a relatively long action sequence or action chunk, reducing the overhead of repeated deliberation [10, 228]. At the other extreme, an agent can repeatedly execute a short action or subgoal, observe the resulting state, and decide what to do next [39, 76]. Neither extreme is universally desirable.

End-to-end planning and execution.. Long-horizon prediction or action chunking allows the policy to execute coherent behavior without repeatedly invoking an expensive high-level model. This is particularly attractive when control must run faster than multimodal reasoning can be performed. However, committing to a long sequence also delays the point at which unexpected environmental changes or execution errors can influence the policy. An action sequence that was reasonable when generated may become incorrect after its first few actions.

Dynamic granularity and horizon. A growing direction is therefore to make the commitment horizon itself adaptive. Recent work explores adaptive action chunking, mixtures of execution horizons, and dynamic execution commitment [20, 85, 109]. Under this view, granularity becomes a decision variable: predictable phases of a task may permit longer commitments, while contact, uncertainty, environmental change, or detected error should trigger shorter horizons and more frequent re-observation. Real-time correction mechanisms similarly aim to incorporate new observations before an entire previously generated chunk has been executed [167].

Fine-grained or step-by-step planning. The alternative is to decompose a task into shorter subgoals or individual actions and repeatedly close the perception–action loop. Hierarchical planning methods provide one mechanism for separating abstract task decisions from lower-level execution [68, 86, 156]. Frequent replanning increases responsiveness but also introduces additional inference cost, may produce temporally inconsistent decisions, and can prevent the policy from exploiting coherent longer-horizon motor structure.

Consider the simple instruction *"reach the cup."* Should the agent generate the reaching trajectory as one complete behavior, explicitly decompose it into approach and grasp-related steps, or choose the execution horizon online depending on visibility, proximity, obstacles, and uncertainty? This example exposes the broader challenge: the appropriate unit of reasoning is not necessarily the appropriate unit of control. A general embodied agent therefore needs to decide not only *what* to do, but also *how long to commit before looking and thinking again*.

5.3 OC3. Verification Architecture: Unified Agent or Dedicated Verifier?

Core question. *Should an embodied agent verify the outcome of its own actions, or should verification be delegated to a separate model or module?*

Closing the loop requires more than generating an action: the system must determine whether the intended change actually occurred. Existing systems use visual feedback to detect task success, identify execution failures, and revise plans [33, 56, 120]. An unresolved architectural question is whether this evaluation should remain a capability of the acting model itself or become an independent component.

Self-verification. A unified agent can use the same foundation model, policy, or VLA that generated the action to interpret subsequent observations and judge whether the action succeeded. Such designs preserve a simple architecture and allow reasoning, execution, observation, and correction to share a common representation. Self-reflection and feedback-based approaches illustrate the broader potential of this paradigm [6, 76, 133]. However, the actor and verifier may also inherit the same blind spots. If an action was selected because the model misunderstood the scene, asking the same model to evaluate the resulting state may reproduce rather than correct the original mistake.

Dedicated verifier. The alternative is to introduce a separate success detector, failure monitor, critic, or reward model [2, 51, 98, 175, 195]. Recent work further explores execution-time verification and action-conditioned failure detection for long-horizon policies [119, 166, 220]. Separating acting from verification can reduce dependence on a single model and allows the verifier to be optimized specifically for detecting subtle failures. The cost is additional training data, inference, synchronization, and system complexity.

More importantly, architectural separation alone does not guarantee independent errors. An actor and verifier trained on similar data or sharing similar visual representations may still fail on the same cases. The key quantity is therefore not only verifier accuracy in isolation, but the *correlation between acting and verification errors*. A central open question is whether increasingly capable foundation models can reliably serve as both actor and judge, or whether robust embodied autonomy will require verification to become an explicitly specialized and partially independent capability.

5.4 OC4. Future Prediction and World Modeling

Core question. *What aspects of the future must an embodied agent predict in order to act reliably?*

World models are commonly motivated by the intuition that an agent should anticipate the consequences of its actions before executing them. However, "predicting the future" can refer to substantially different computational objectives. Early visual prediction approaches explicitly forecast future observations for robotic planning [35, 40], while more recent world models support latent imagination and model-based control [59, 235]. The open question is whether an embodied agent actually needs a high-fidelity simulation of everything that may happen, or only the subset of future information that changes its decision.

Generative prediction. One approach predicts future images, video, or other sensory observations under candidate actions. Such predictions provide a rich interface because many downstream questions can in principle be answered from the generated future. They may also expose spatial and interaction consequences that are difficult to specify symbolically. However, producing perceptually detailed futures can devote model capacity to aspects of the scene that are irrelevant to action selection, while prediction errors can compound over long horizons.

Latent dynamics modeling. A second approach predicts transitions in a compact latent or structured state rather than reconstructing every future pixel [59, 176]. The representation can focus on information useful for control and make multi-step prediction substantially cheaper. The difficulty is ensuring that the latent state retains precisely those details that become important for downstream decisions, especially when task requirements change.

Task- or action-oriented prediction. A third possibility is to predict only decision-relevant consequences: whether an action is feasible, whether contact will occur, how progress or risk will change, or which candidate action is preferable [87, 105, 226]. World-model representations are also increasingly used directly for execution-time failure prediction rather than full future reconstruction [119, 220].

This suggests that the central challenge may not be to build the most visually accurate simulator, but to identify the *minimal predictive representation sufficient for good decisions*. Different tasks may require different predictive abstractions: geometry for navigation, contact and force for manipulation, or success and risk for high-level planning. Determining what an embodied agent actually needs to know about the future remains an open problem.

5.5 OC5. Acting Under Uncertainty

Core question. *When its current belief is uncertain, should an embodied agent act, actively acquire more information, or stop and seek external assistance?*

Uncertainty is unavoidable in embodied interaction. An object may be occluded, an instruction may be ambiguous, a detector may return conflicting evidence, or an action may have an uncertain physical consequence. Importantly, uncertainty does not imply that the agent should always stop. Robust autonomy requires deciding *what to do with uncertainty*.

Act under uncertainty. When uncertainty is low or the cost of an error is small, the agent may continue using its current belief. This requires uncertainty estimates that are meaningfully related to execution risk rather than merely reflecting model confidence. Recent work studies uncertainty-aware robot planning and failure detection [99, 108, 212]. A major difficulty is calibration under distribution shift, where a model may be highly confident precisely because it fails to recognize an unfamiliar situation.

Active information acquisition. When additional information is valuable, the agent can act specifically to reduce uncertainty: moving the camera, changing viewpoint, inspecting an occluded region, re-observing the scene, or executing a diagnostic action before committing to the task. Active perception has consequently re-emerged as an important component of foundation-model-based embodied systems [122, 147, 164]. Related work explicitly treats exploration as a process that should continue until sufficient confidence has been obtained [159]. The challenge is that information gathering itself consumes time and may alter the physical environment, so the value of a new observation must be weighed against its cost.

Ask, abstain, or defer. Finally, an agent may decide that autonomous execution is not justified. It can ask a human to clarify an instruction, request assistance, abstain from a high-risk action, or transfer control to another agent or human operator. Methods that explicitly align planning uncertainty with help-seeking demonstrate the importance of this capability [158], while human-in-the-loop and modular safety architectures provide broader mechanisms for external intervention [93, 141].

The ultimate problem is therefore not uncertainty estimation alone, but *uncertainty-conditioned control*. A general embodied agent requires a meta-policy over at least three alternatives: *act with the current* *belief*, *acquire information*, or *ask, abstain, or defer*. The correct choice depends jointly on uncertainty, the expected information gain of further observation, the reversibility of the candidate action, and the cost of failure. Learning this trade-off across diverse tasks and embodiments remains a central challenge for reliable embodied autonomy.

6 Conclusion

In this survey, we examined multimodal embodied agents through a closed-loop, capability-centric perspective. Rather than organizing the literature primarily by model family, architecture, task, or robotic platform, we introduced PAPAV, Perceive, Anticipate, Plan, Act, and Verify, as a common coordinate for comparing how digital agents, robotic systems, and multimodal embodied agents realize the functions required to complete an interaction loop. This perspective allowed us to preserve the strengths of both agent and robotics research while focusing directly on what changes when an agent's decisions must be grounded and executed in the physical world.

Across the five capabilities, a consistent picture emerges. Physical embodiment does not fundamentally replace the interaction loop; instead, it changes the conditions under which the loop must be maintained. Partial observability, limited simulatability, irreversibility, real-time coupling, and unverifiability reduce the amount of structure that can be assumed in advance and shift greater responsibility to the running system. As a result, persistent state estimation, uncertainty-aware prediction, constraint formulation and revision, management of physical commitments, and evidence-based verification become increasingly important to reliable autonomy. Our benchmark analysis further shows that progress in end-to-end task performance does not necessarily imply that these capabilities are individually understood or reliably evaluated, motivating more diagnostic evaluation of both capability outputs and their influence on subsequent decisions.

Despite rapid advances in multimodal foundation models, world models, generalist robot policies, and agentic systems, building robust MMEAs remains an open systems problem. Important questions remain around how persistent beliefs should be represented, how planning and execution granularity should adapt during interaction, how verification should be organized, what aspects of the future should be predicted, and how uncertainty should govern whether an agent acts, gathers more information, or seeks assistance. Addressing these questions will require not only stronger individual models, but also better mechanisms for keeping perception, prediction, planning, execution, and verification coordinated throughout physical interaction. We hope that the PAPAV framework provides a useful common vocabulary for studying these relationships and for guiding the design and evaluation of more capable, reliable, and general multimodal embodied agents.

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